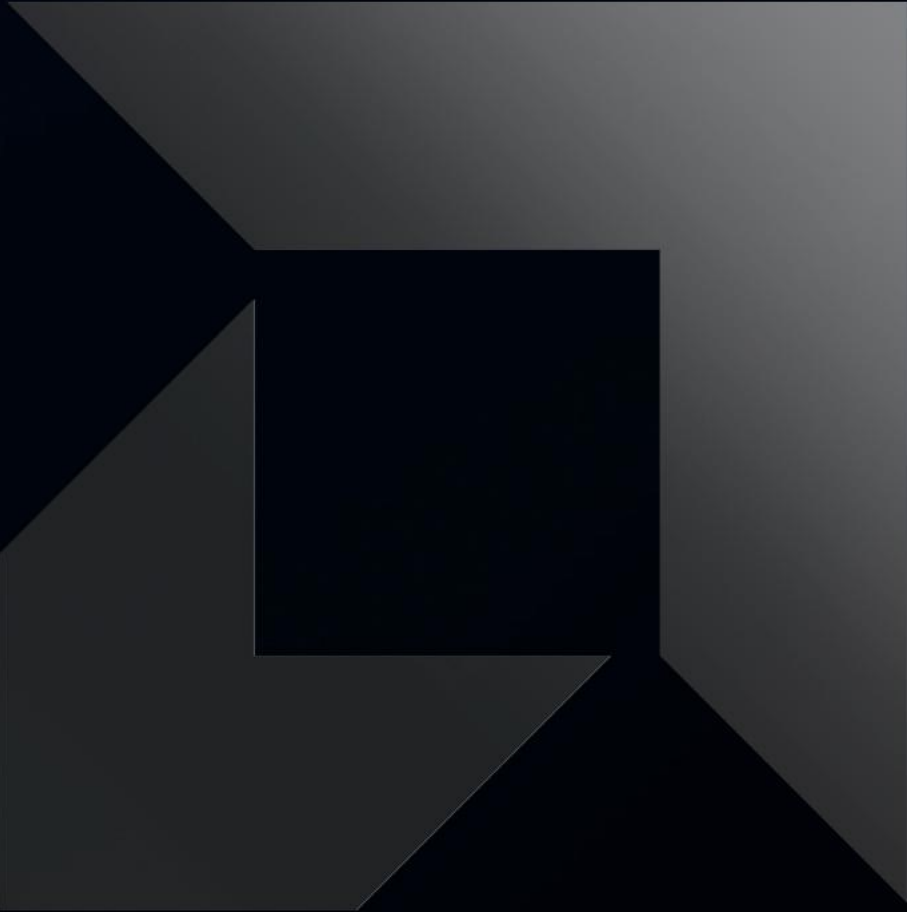




Reference Cluster Design Architecture MI300X / MI325X

1K GPU to 8K GPU

Version 4.1 – February 2026

Cluster - 1024 GPU to 8192 GPU (128 to 1024 Nodes) *

Cluster Size	1024 to 8192 GPU
Platforms	Dell XE9680 Lenovo SR685a V3 SMCI AS-8125GS
OS	Ubuntu 22.04 (or above)
Linux Kernel	5.15 – 6.80
ROCm	6.3.3 (or above)

*Genericized BoM – removing device specifics including dependencies

Storage Type	
Local Storage	1.6 TB (or greater)
Utility Storage	Pure, Vast, RYO
Bulk Storage	Pure, Vast, WekaIO
Scratch Storage	Vast, DDN, WekaIO, Hammerspace
Archive/Object Storage	S3 Compatible

Network - 1024 GPU to 8192 GPU (128 to 1024 Nodes) *

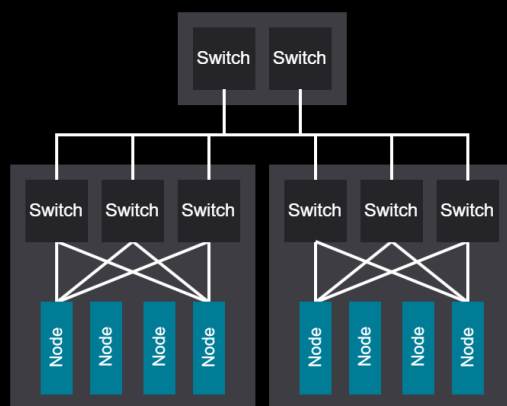
Backside Network Topology	2 Tier Rail Optimized / Fat Tree
NIC	Pollara 400 BCM957608 (Thor2)
Switch	Arista, Dell, Juniper, Cisco (TH 4/5, Scheduled fabrics)
Network OS	SONiC, Junos, EOS, IOS, DriveNets
Subscription Ratio	1:1.16=16% Undersubscribed (AMD recommendation)
Optics	Vendor ACL/HCL
Fabric	ROCEv2 Ethernet

Frontside Network Segment	Adapter Recommended
All-in One Network	Ethernet 100GbE 2-port QSFP28 Adapter
Storage Network (Optional)	Ethernet 100GbE 2-port QSFP28 Adapter
Virtualization Network (Optional)	Ethernet 100GbE 2-port QSFP28 Adapter
Host In-Band	Ethernet 10/25GbE 4-Port SFP28 Adapter
BMC OOB Mgt	1G Copper

* Genericized BoM – removing device specifics including dependencies

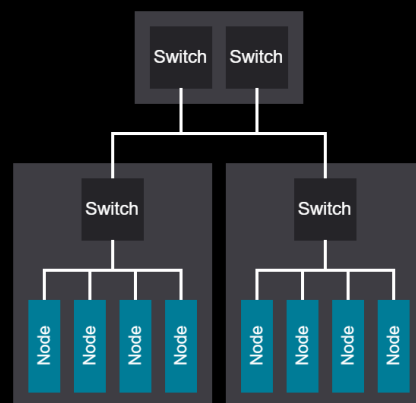
Cluster Network

Basic Network Topologies



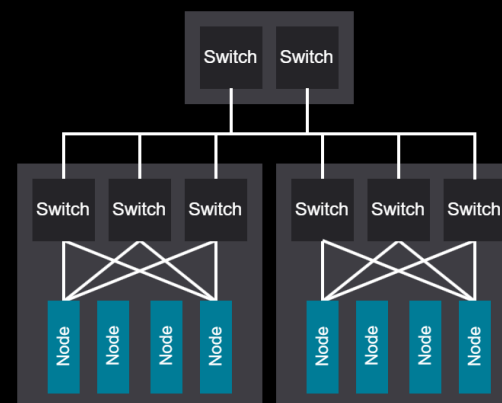
2-Tier Rail Network

- Enables large scalable unit sizes for large jobs or replica sizes
- Efficient for workloads favoring ring-based collectives
- Higher infrastructure costs



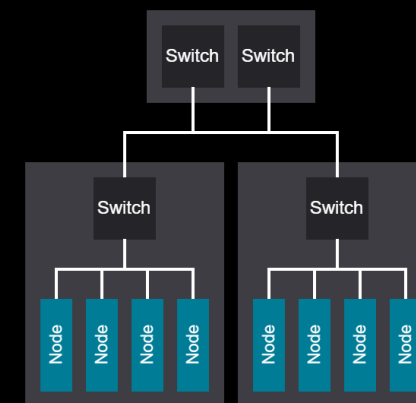
2-Tier Tree Network

- Efficient for small workloads or replicas
- Easy to add capacity with proper planning
- Potentially allows for lower infrastructure costs
- Limited blast radius compared to rail networks



3-Tier Rail TH5/J3 Network

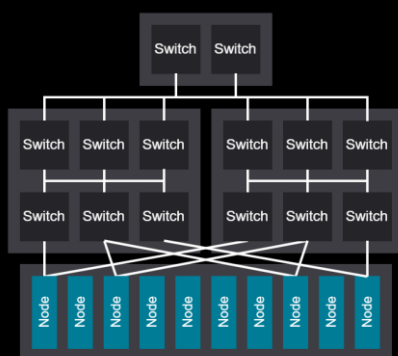
- Spine switches replaced with a 2-tier Jericho3-AI/Ramon3 fabric for increased maximum cluster size
- Deep buffers and scheduled fabric aid greatly with congestion issues in large clusters at a small latency penalty



3-Tier Tree TH5/J3 Network

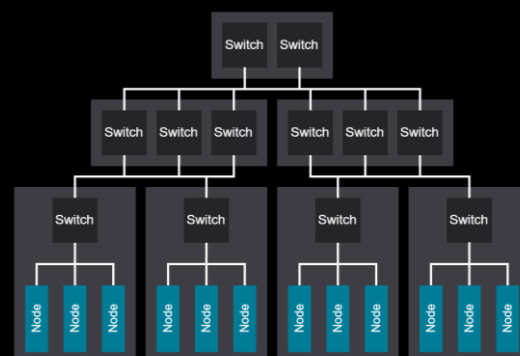
- Same benefits from switch to scheduled spine fabric as with rail, but retains the primary characteristics of tree networks

Basic Network Topologies Continued



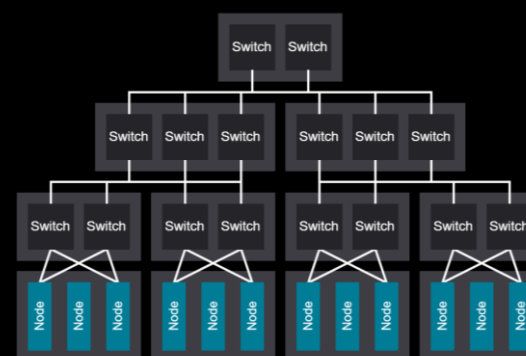
3-Tier Rail Optimized Network

- Allows for massive scalable unit sizes
- Best ring-based collective performance at scale; at the cost of poor any-any performance



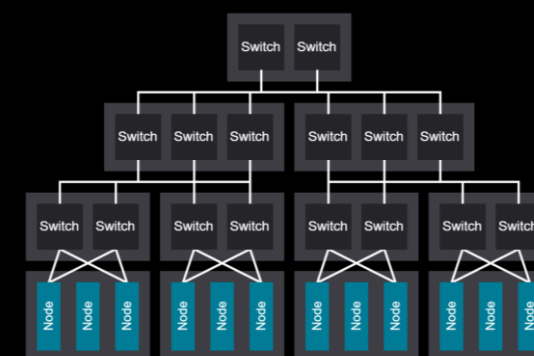
3-Tier Tree Network

- Allows for massive cluster sizes
- Best any-any performance at scale
- Suitable for a “campus style” deployment



3-Tier Hybrid Rail Network

- Allows for massive cluster sizes with large scalable units
- Favors ring-based collectives, but does not sacrifice significant any-any performance on large jobs
- Suitable for a “campus-style” deployment

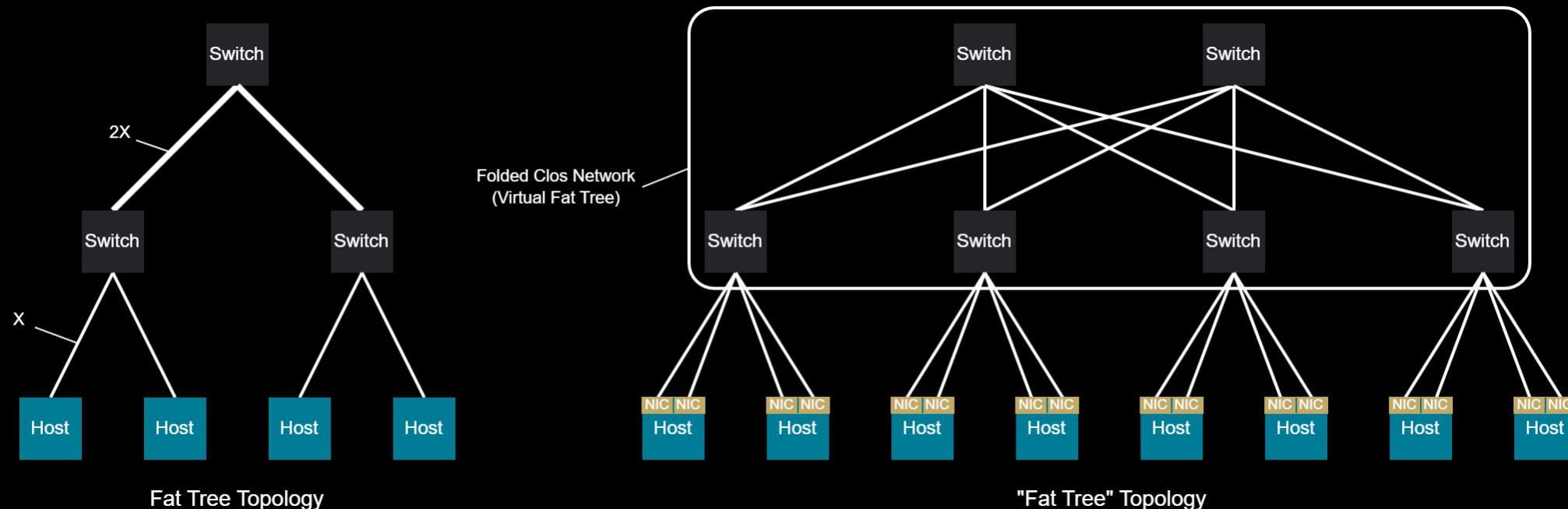


3-Tier Fully Scheduled Rail Network

- Medium sized scalable units
- Excellent congestion performance due to deep buffers and scheduled fabric
- Technical limitations limit cluster size to ~32K GPUs in recommended configuration

Tree and Rail

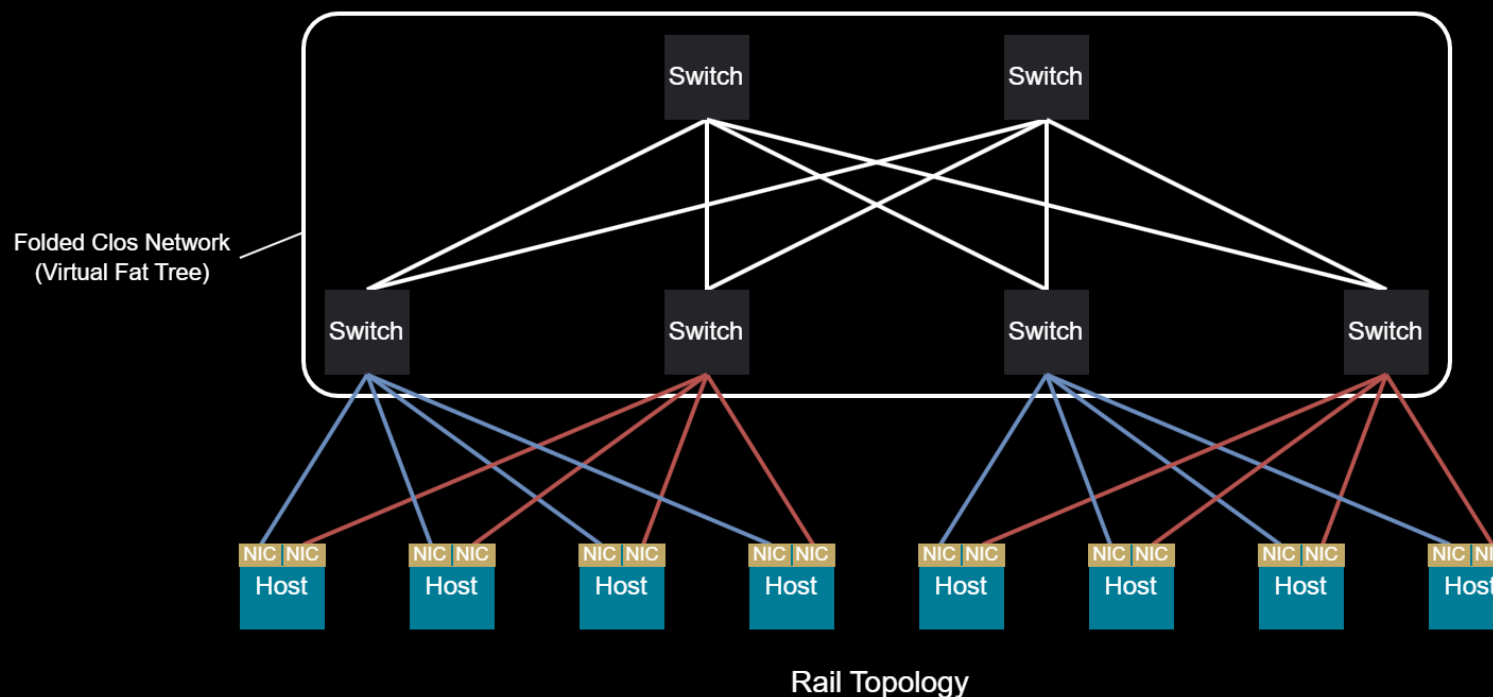
Fat Tree Terminology In Cluster Design



The canonical fat tree topology is a network concept where a switch's connection to upstream peers has at least parity bandwidth with the total aggregate bandwidth of its downstream connections. This causes links between switches to become “fatter” as they get closer to the core.

The “fat tree” topology for AI/ML clusters instead refers to how a host is connected to its upstream switches; in this case all host NICs terminate on the same switch. It can also be considered 1-rail network. The network itself is generally a 3-stage or 5-stage folded Clos network due the fixed radix of network switches.

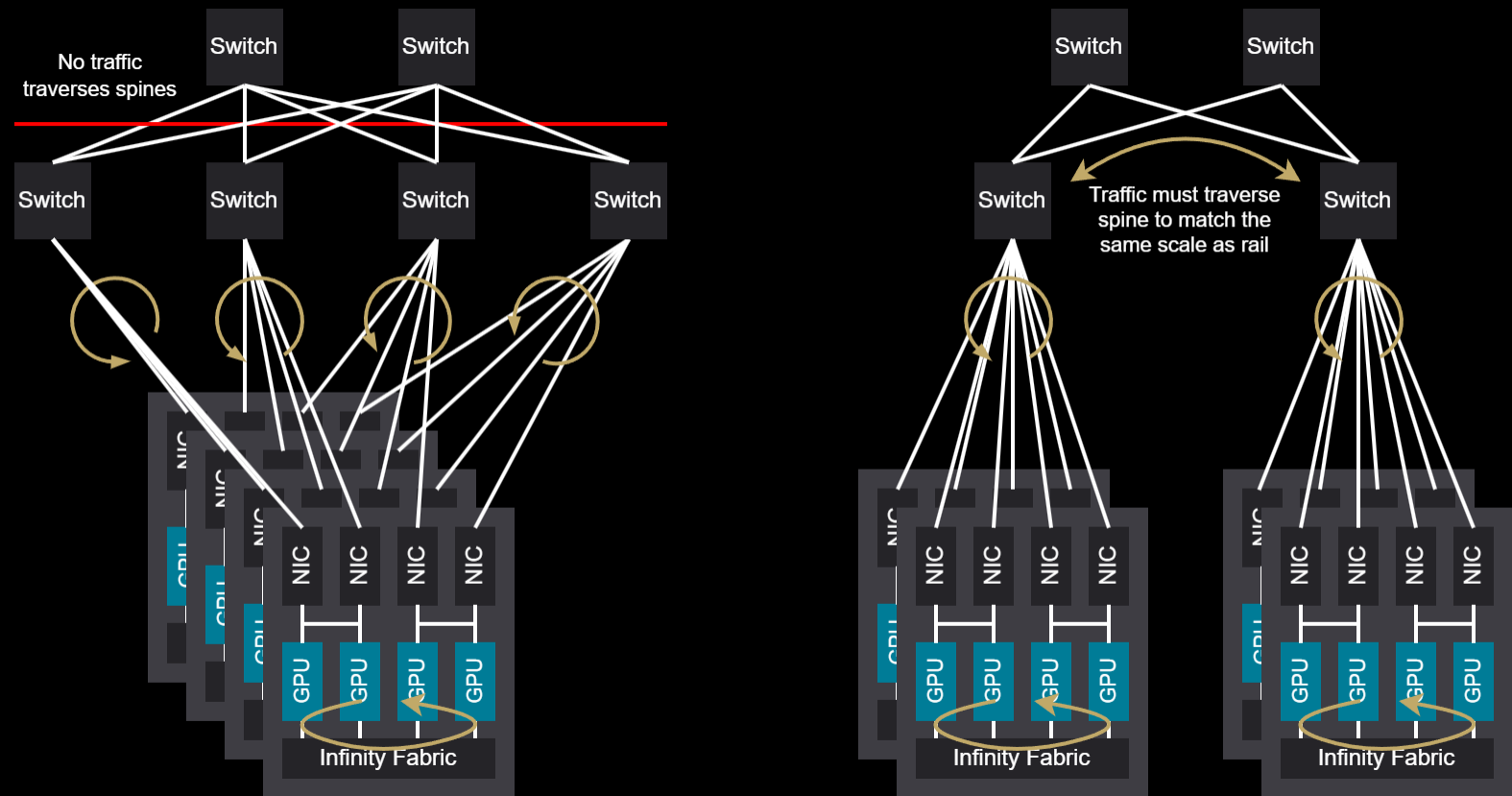
Rail Networks In Cluster Design



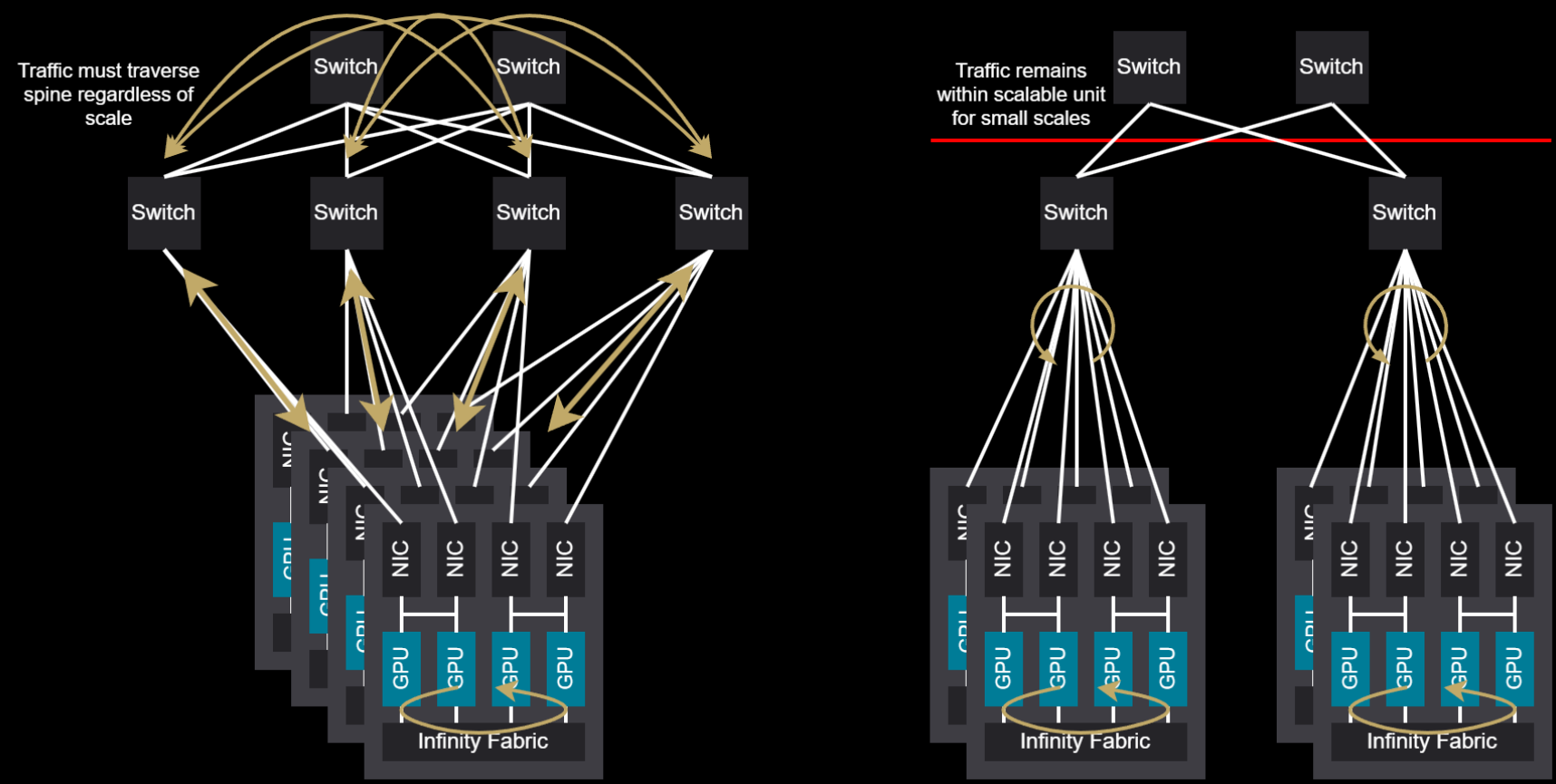
Rail networks leverage the same folded Clos network as tree networks, but host connections are instead aggregated onto switches based on NIC rank. These shared ranks are referred to as rails and allow the network to provide preferential latency for connections which share the same rail. The downside to this design is any traffic which needs to cross rails/ranks must traverse either the network spine layer, or Infinity Fabric (PXN).

The above example shows an example 2-rail network, with the rails colored blue and red to differentiate them.

Rail Enables Larger Single Hop Ring Domains

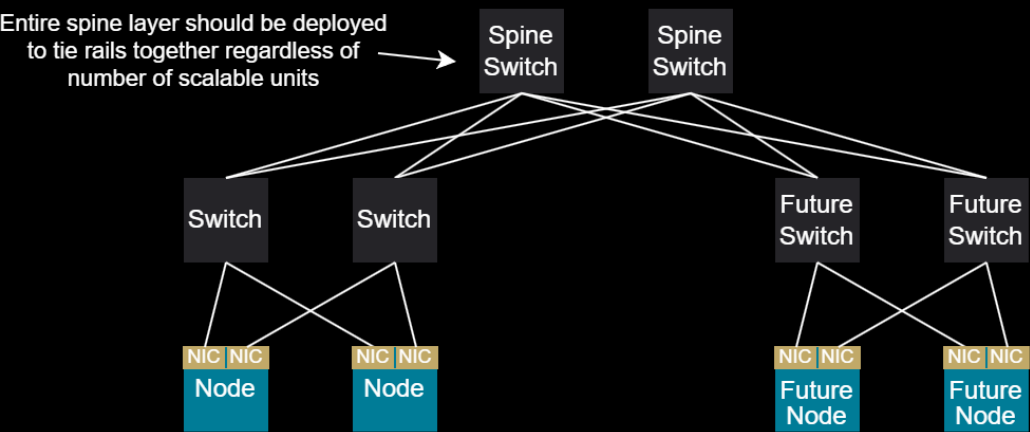
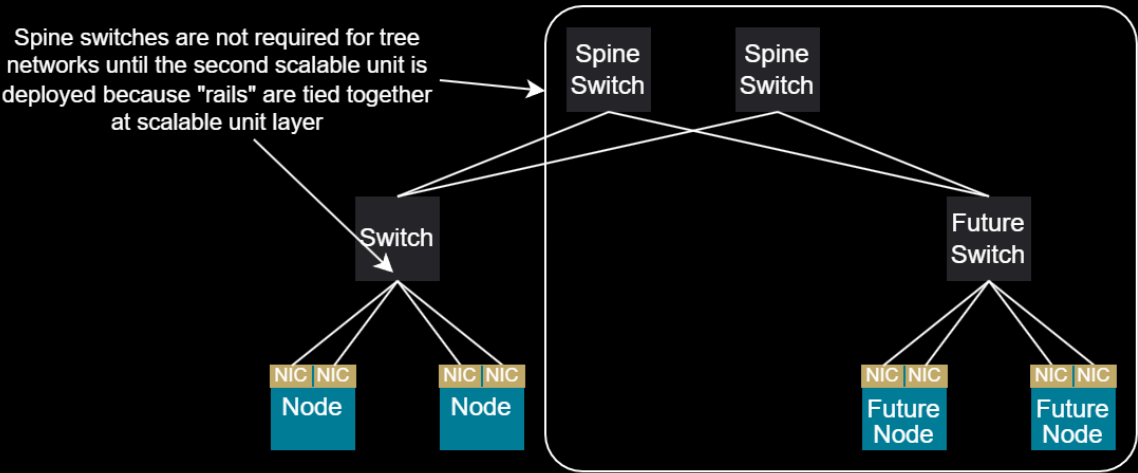


Tree Handles Cross-Rank Traffic Better

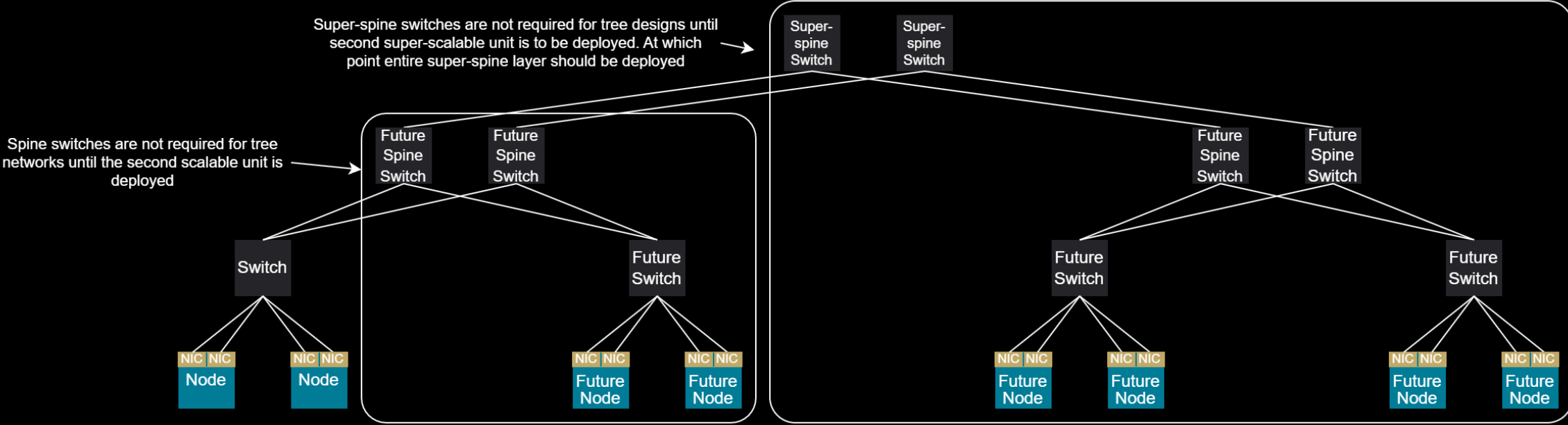


Scaling Networks

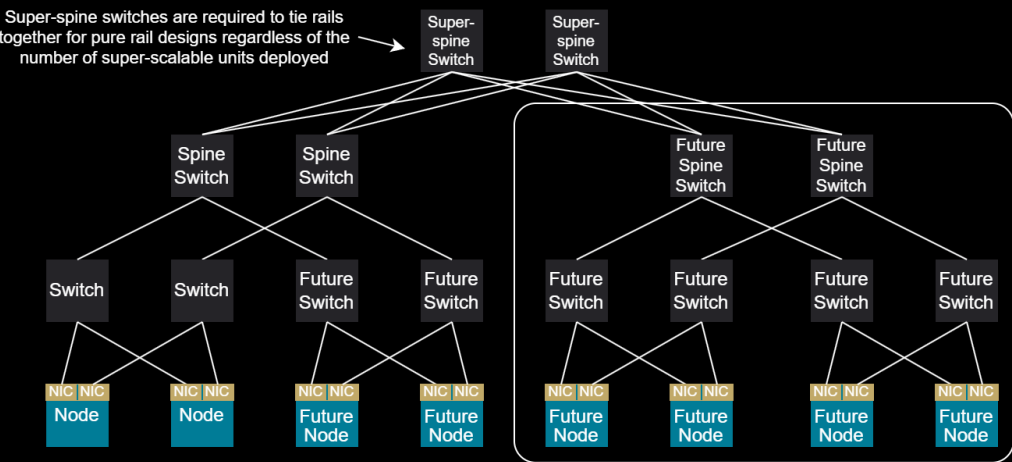
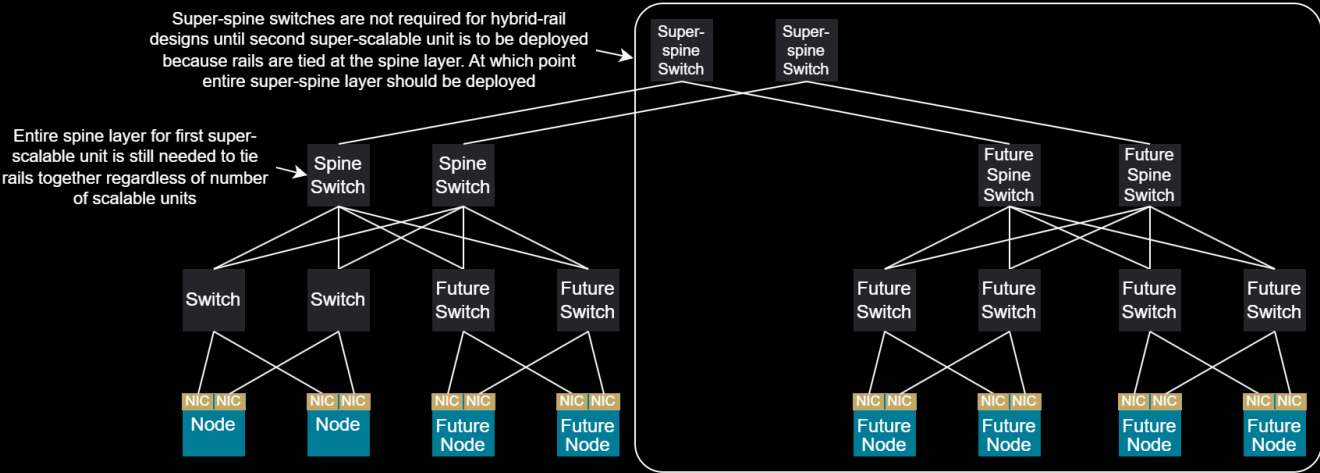
Cluster Backend Deployment Strategy for 2-tier Networks



Cluster Backend Deployment Strategy for 3-tier Tree Networks



Cluster Backend Deployment Strategy for 3-tier Rail Networks



Network Subscription

Subscription is the relationship between what is provided by the upstream network and what is required by the downstream network in demand side.

It is typically represented as a ratio:

$$\text{Downstream Demand} : \text{Upstream Capacity}$$

1:1 subscribed network would be equal downstream capacity to upstream capacity.

Example: 1:1.16 = (1 downstream demand : 1.16 upstream capacity. In this example there is .16 more upstream capacity)

Or as a percentage:

$$\text{Subscription Rate} = \frac{\text{Downstream Demand}}{\text{Upstream Capacity}}$$

80% subscription ratio could be referred to as “20% undersubscribed”, or a 120% subscription ratio could be referred to as “20% oversubscribed”.

Network Design Examples

Network Topology Design Notations:

Design note:

Designs included are based on scheduled fabrics switch type (Arista, Ciena, Nokia) or 51.2T switch type (Arista, Cisco, Dell, Juniper)

Vendors/switch models vary for port count and features – please consult desired vendor port count directly to confirm.

Scalable Units/PODs note:

Diagrams presented are designed around a Scalable Unit or POD – which can determine overall network end to end latency and AI use cases.

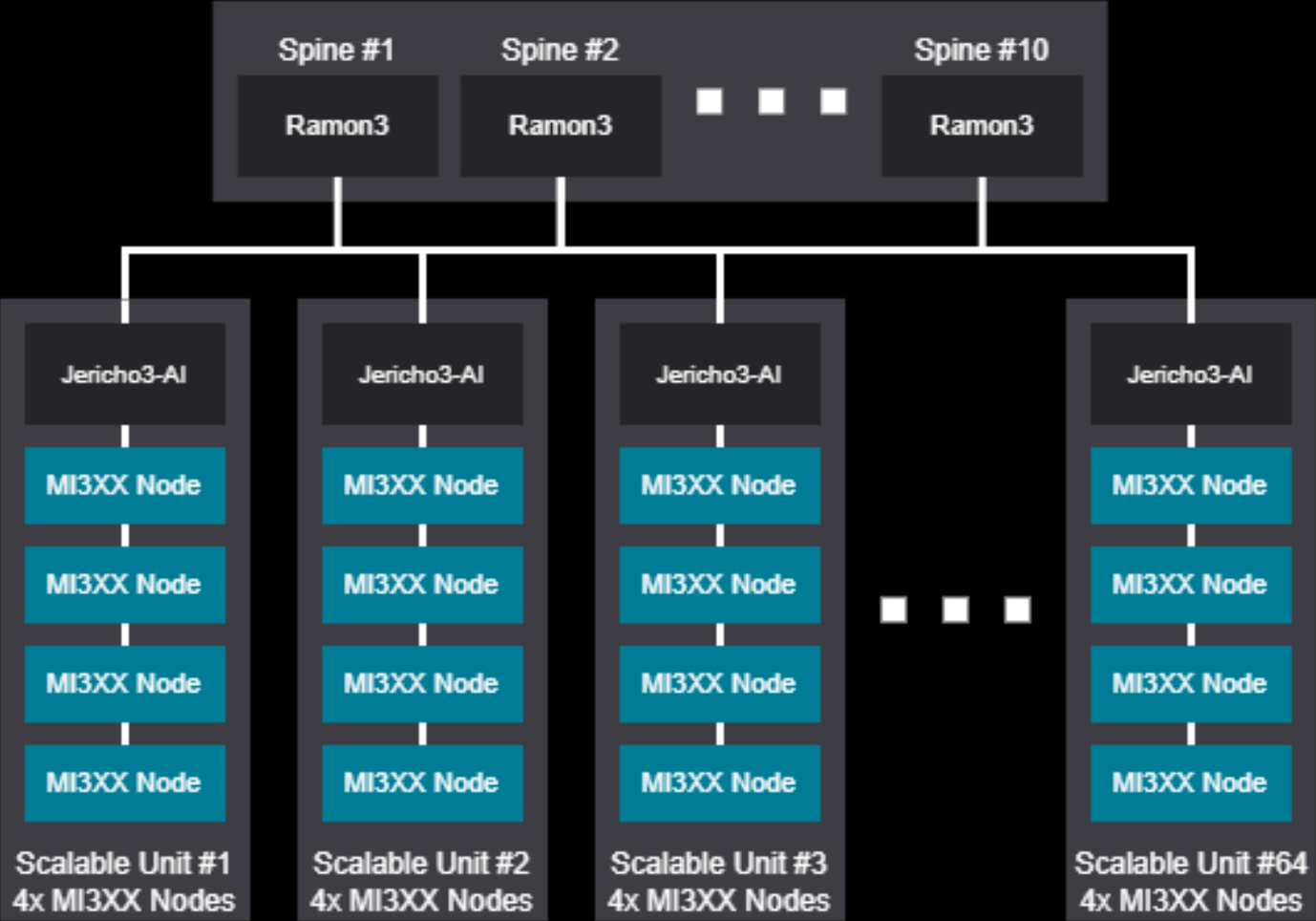
Certain ML/AI workloads may require change of scalable unit size. Please consult with AMD Architecture as required.

2K GPU Topology Design Examples – Scheduled Fabrics

Network Diagram - 2048 GPU (256 Nodes)

Tree Design

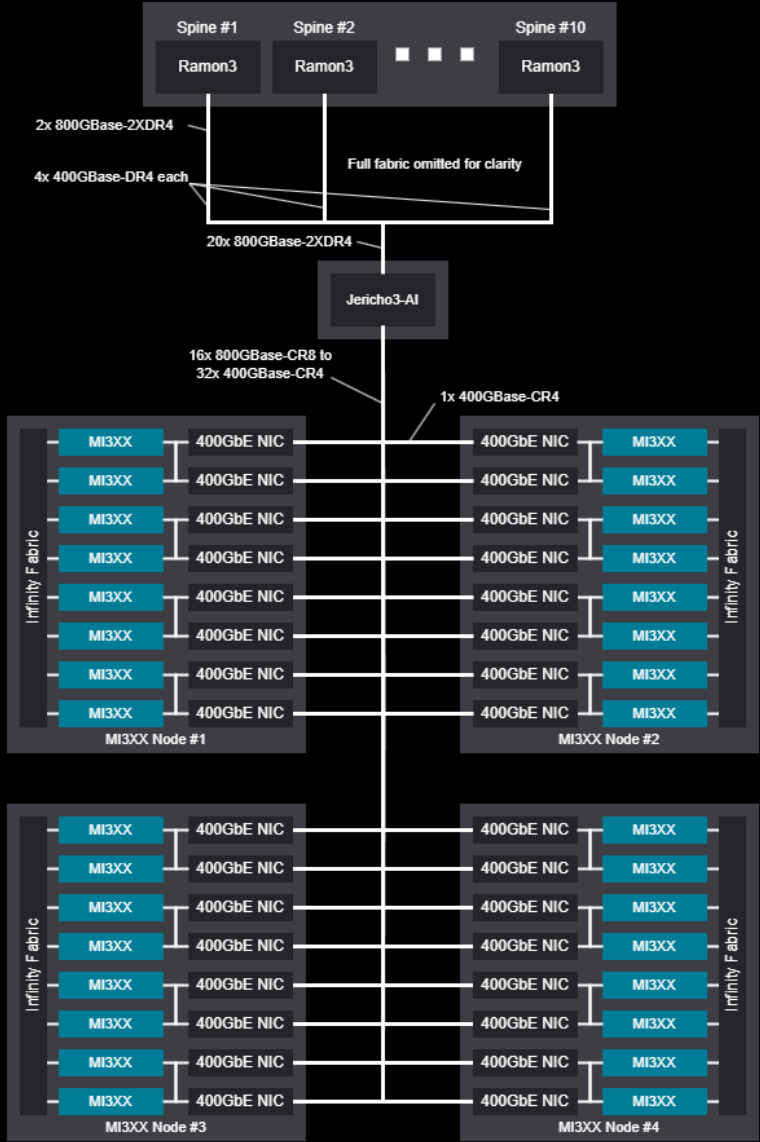
Jericho/Ramon



Network Diagram - 2048 GPU (256 Nodes)

Tree Scalable Unit

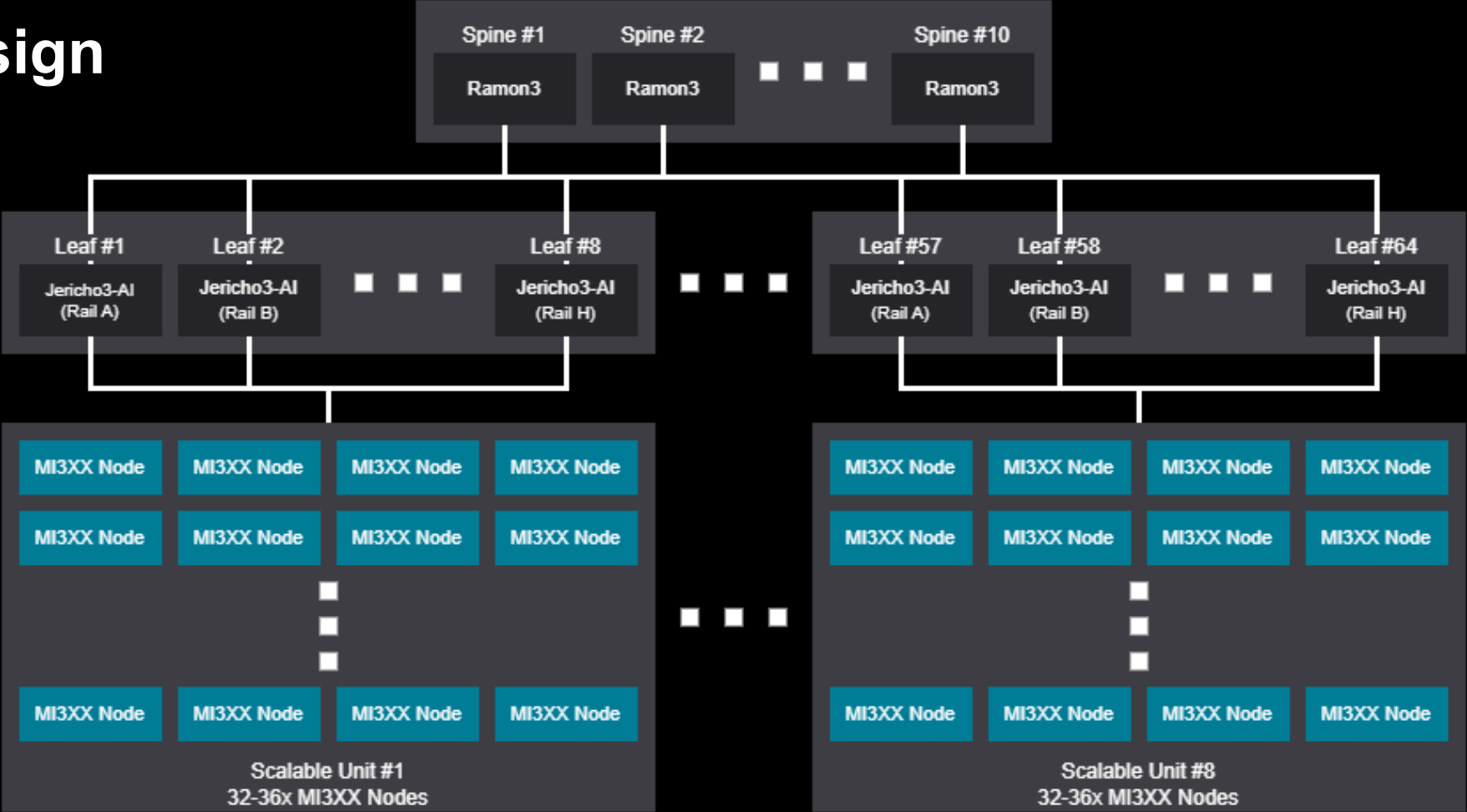
Jericho/Ramon



Network Diagram - 2048-2304 GPU (256-288 Nodes)

Rail Design

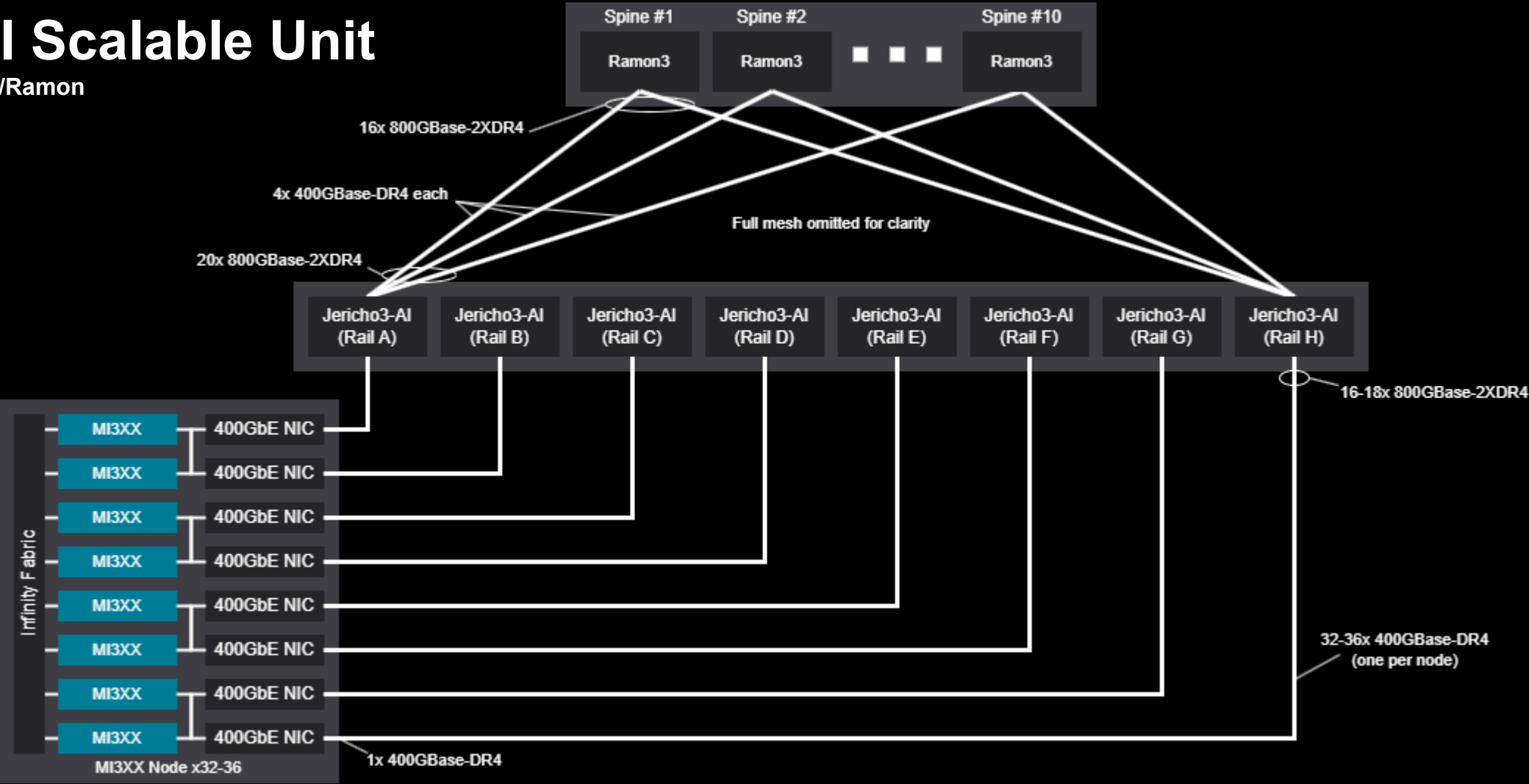
Jericho/Ramon



Network Diagram - 2048-2304 GPU (256-288 Nodes)

Rail Scalable Unit

Jericho/Ramon



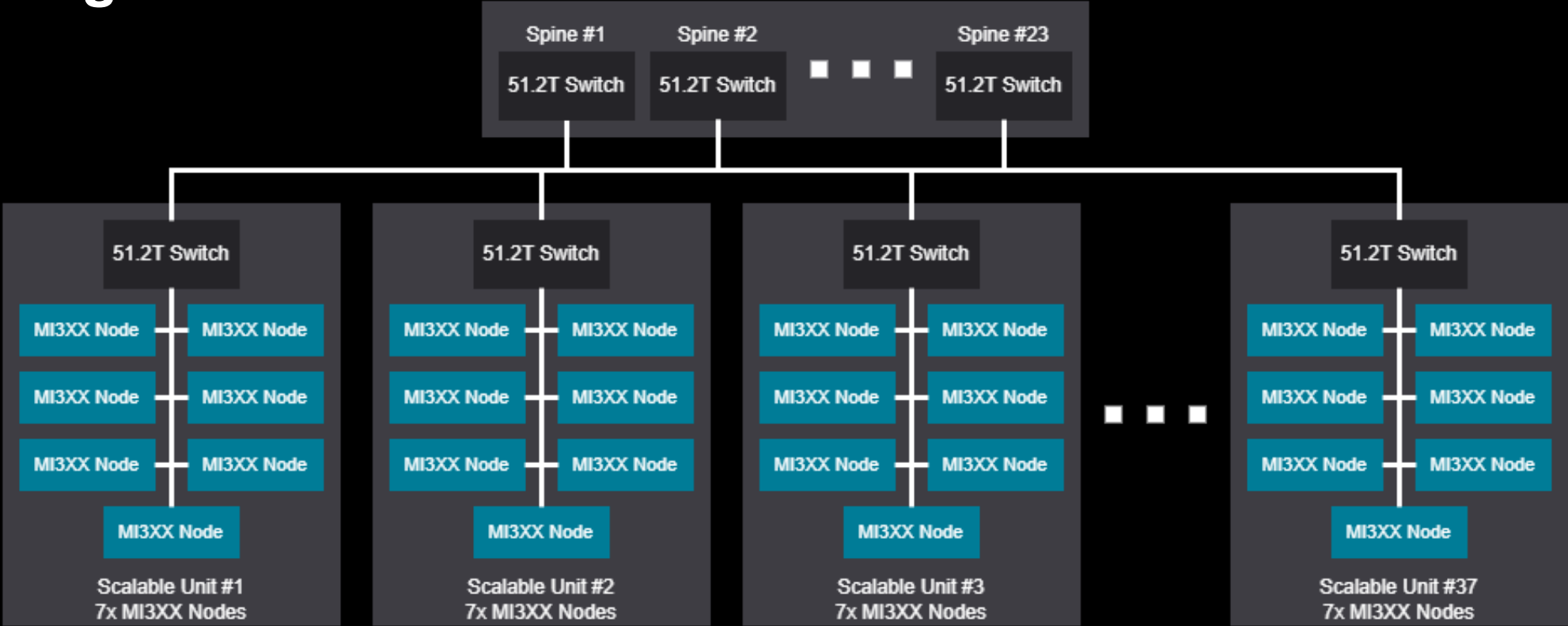
2K GPU Topology Design Examples

51.2T

Network Diagram - 2072 GPU (259 Nodes)

Tree Design

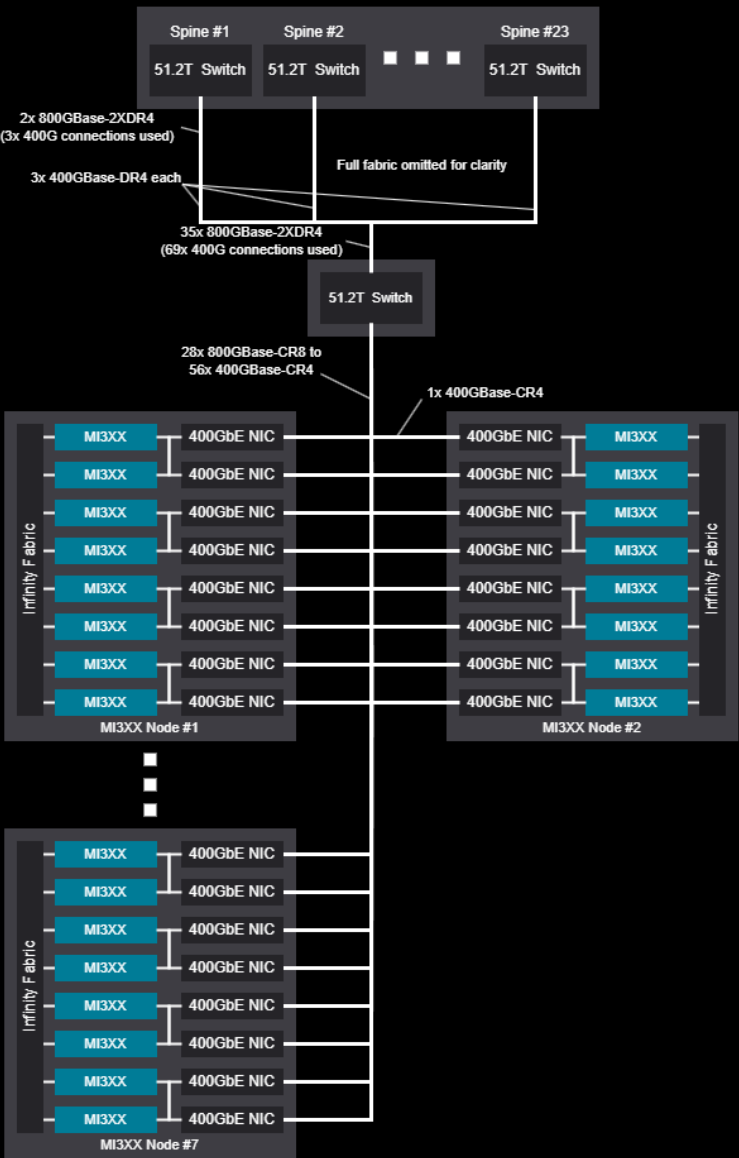
51.2T



Network Diagram - 2072 GPU (259 Nodes)

Tree Scalable Unit

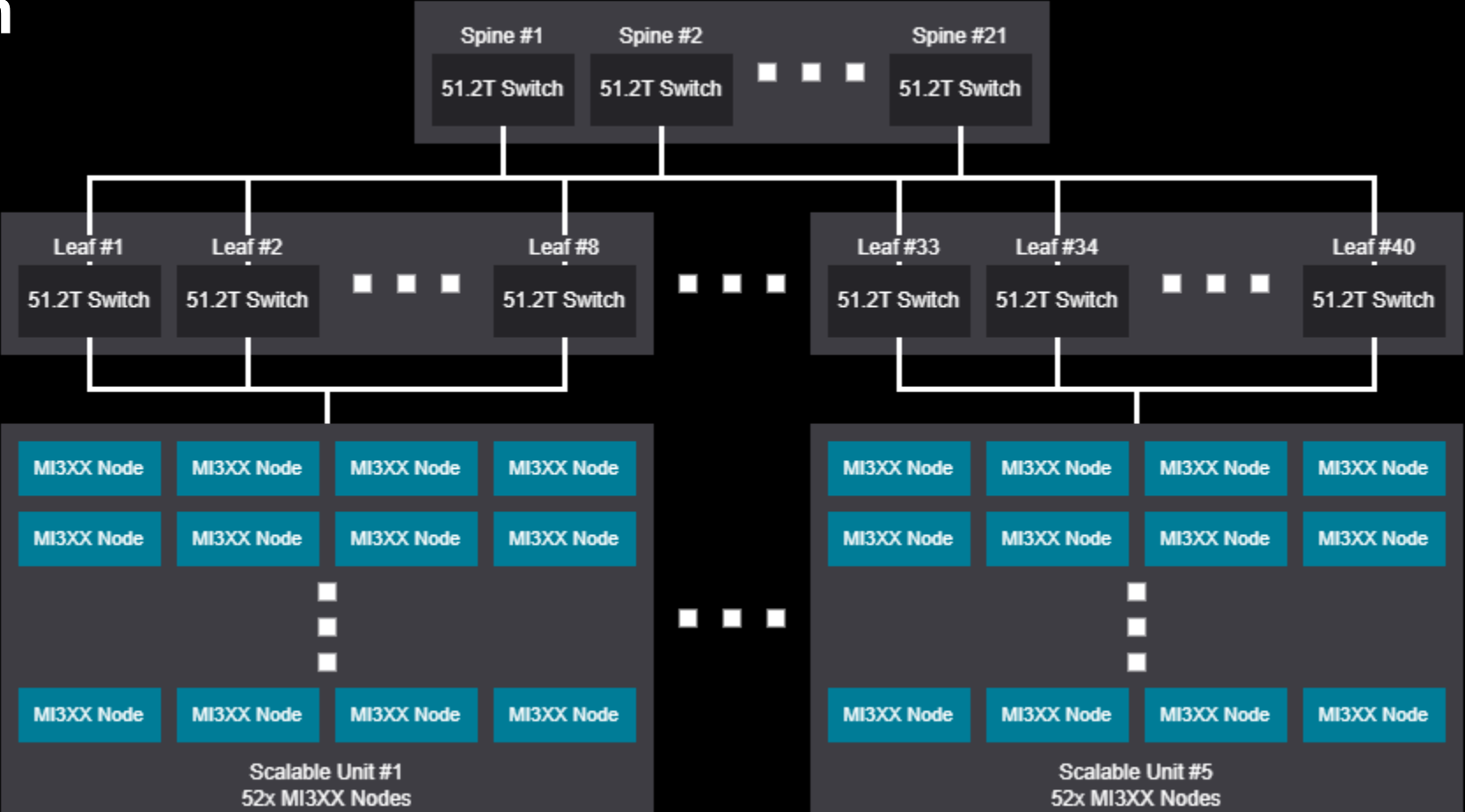
51.2T



Network Diagram - 2080 GPU (260 Nodes)

Rail Design

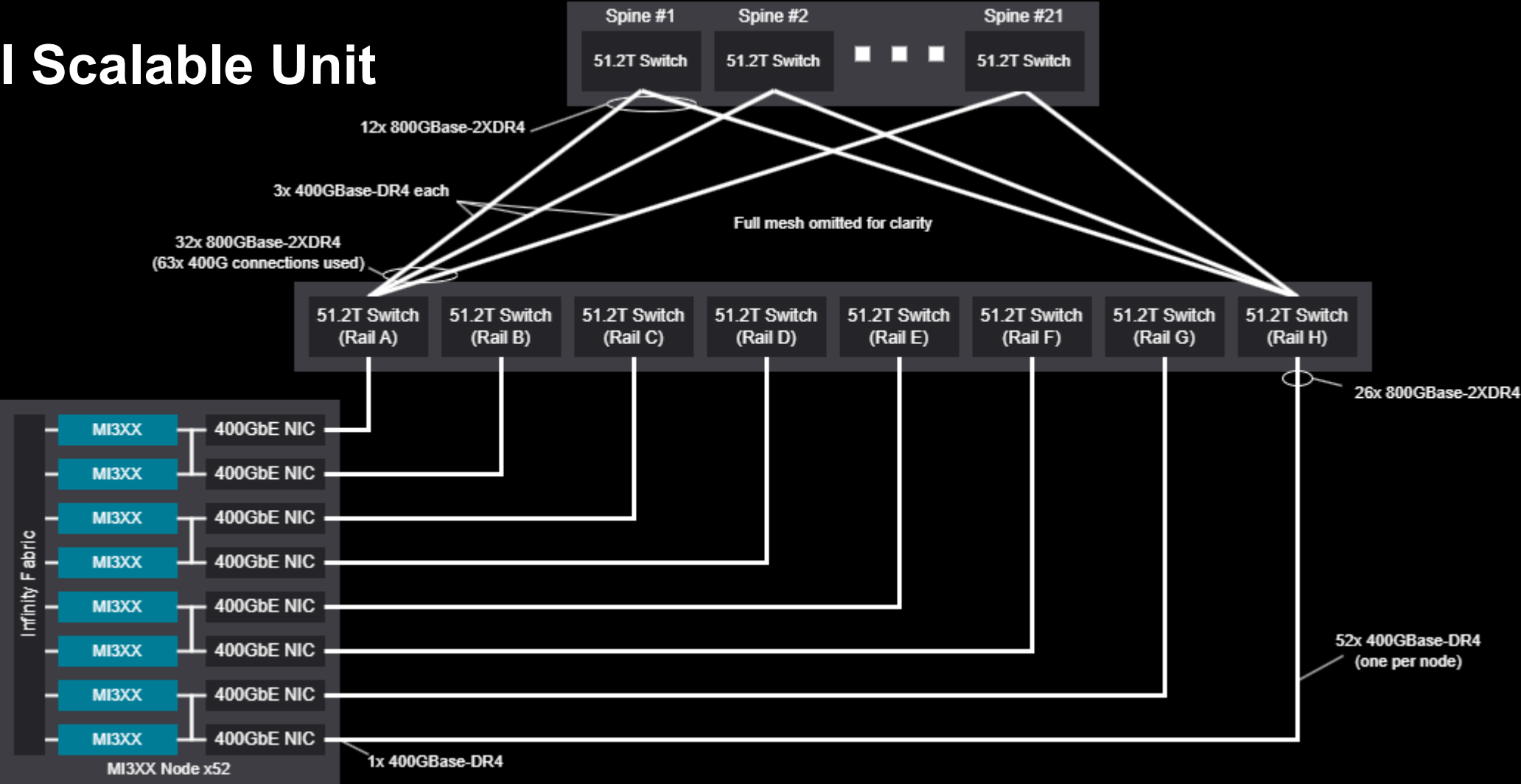
51.2T



Network Diagram - 2080 GPU (260 Nodes)

Rail Scalable Unit

51.2T

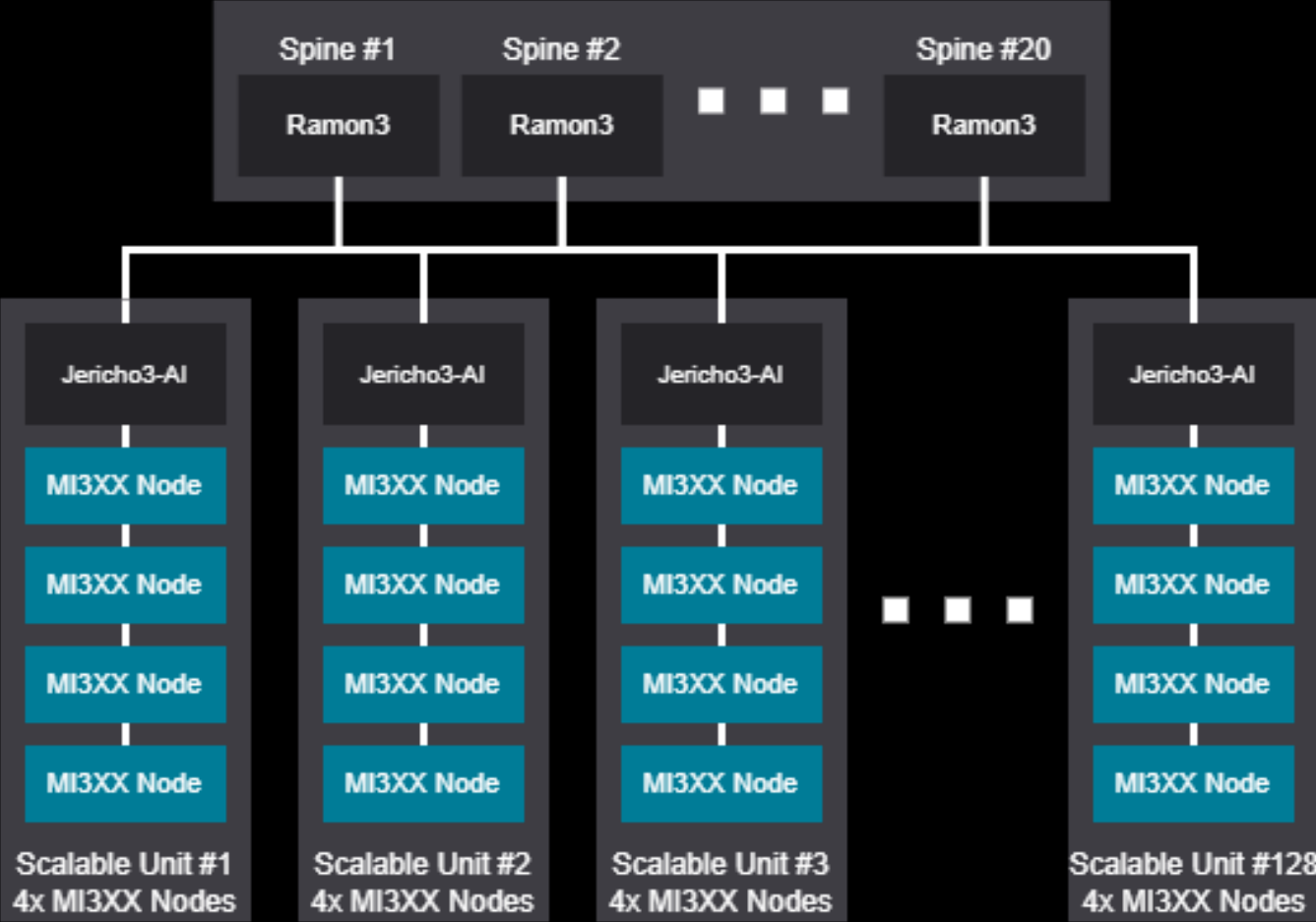


4K GPU Topology Design Examples – Scheduled Fabrics

Network Diagram - 4096 GPU (512 Nodes)

Tree Design

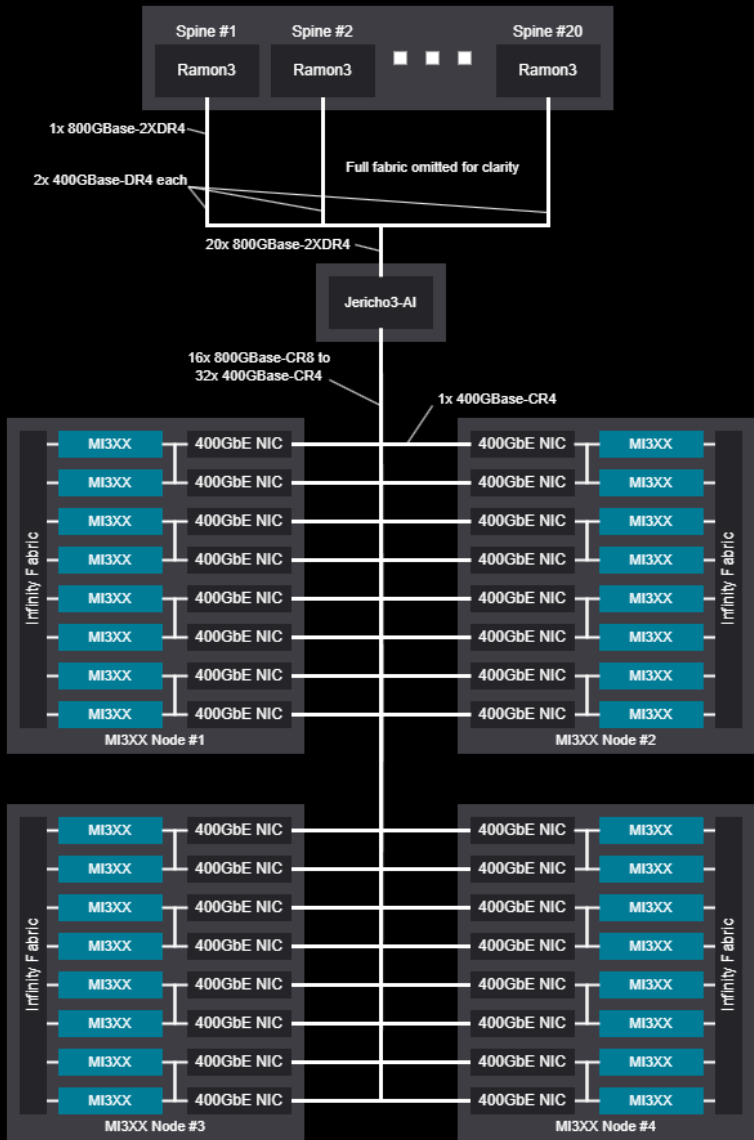
Jericho/Ramon



Network Diagram - 4096 GPU (512 Nodes)

Tree Scalable Unit

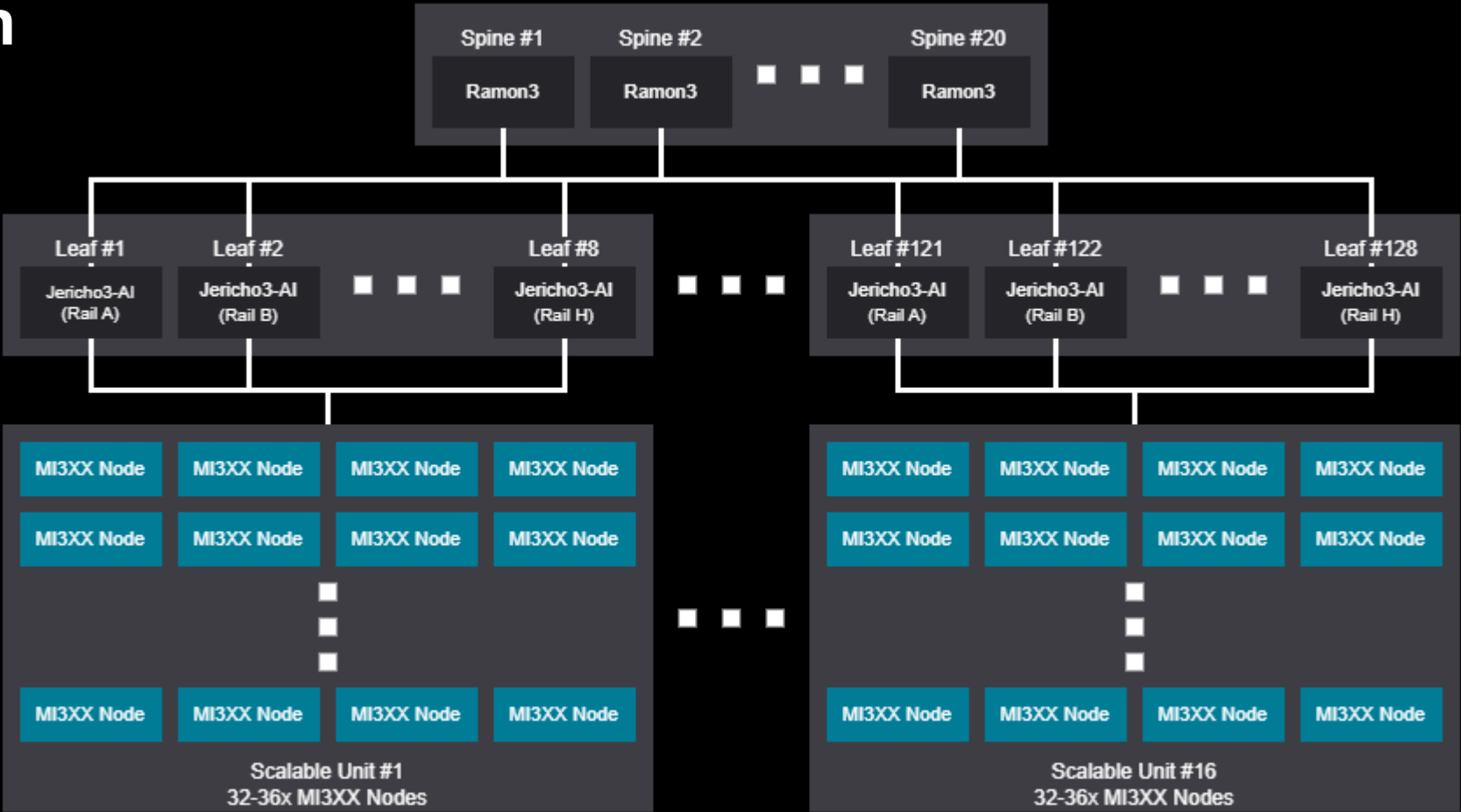
Jericho/Ramon



Network Diagram - 4096-4608 GPU (512-576 Nodes)

Rail Design

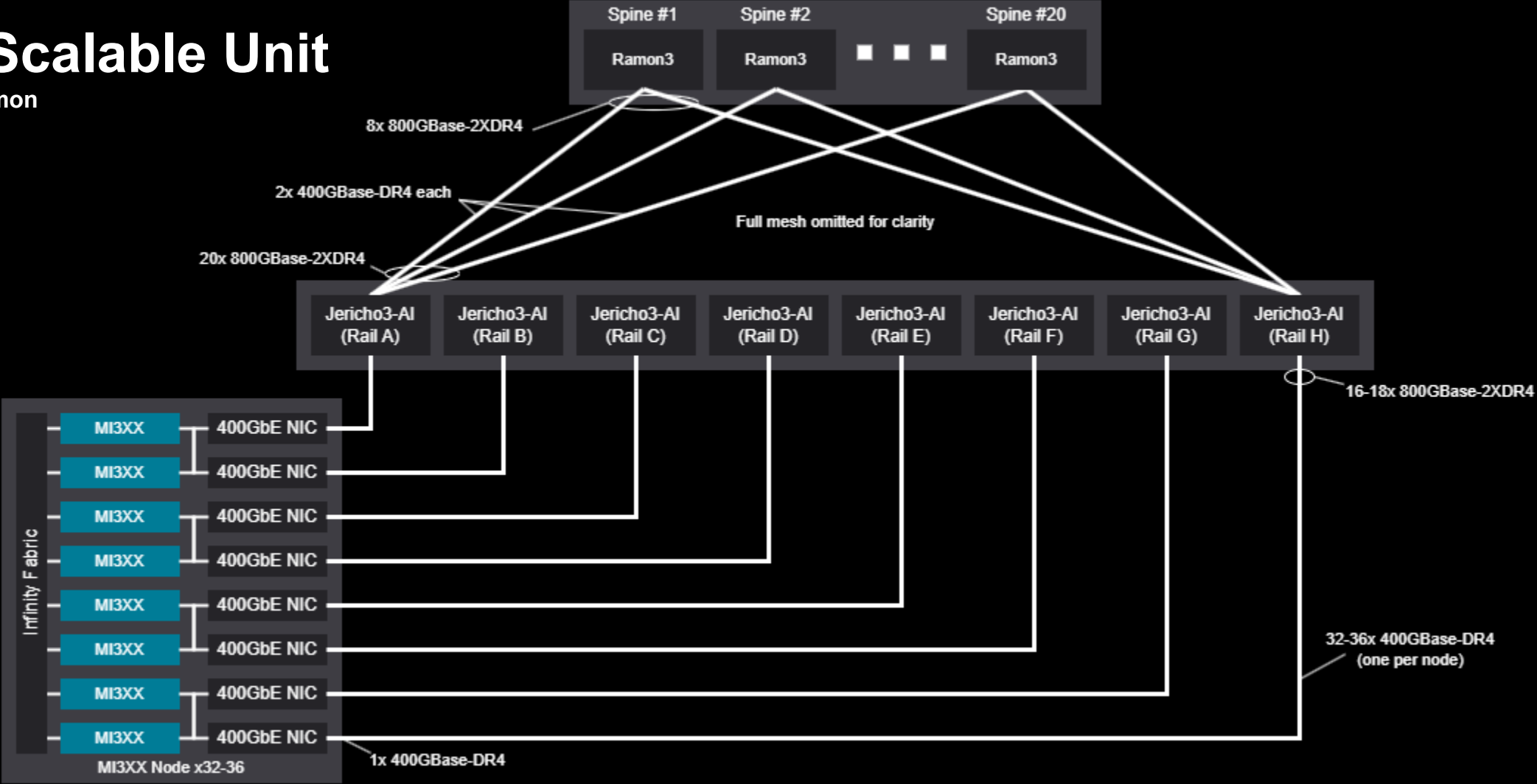
Jericho/Ramon



Network Diagram - 4096-4608 GPU (512-576 Nodes)

Rail Scalable Unit

Jericho/Ramon



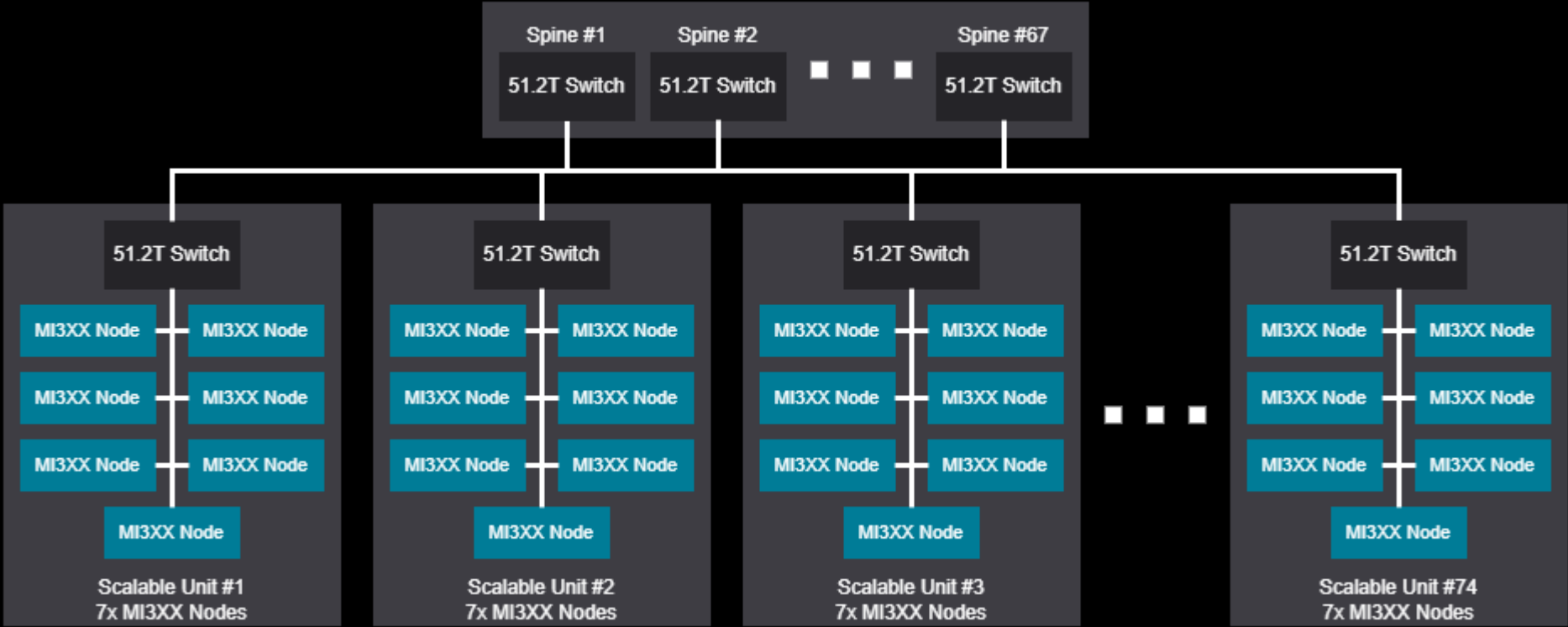
4K GPU Topology Design Examples

51.2T

Network Diagram - 4144 GPU (518 Nodes)

Tree Design

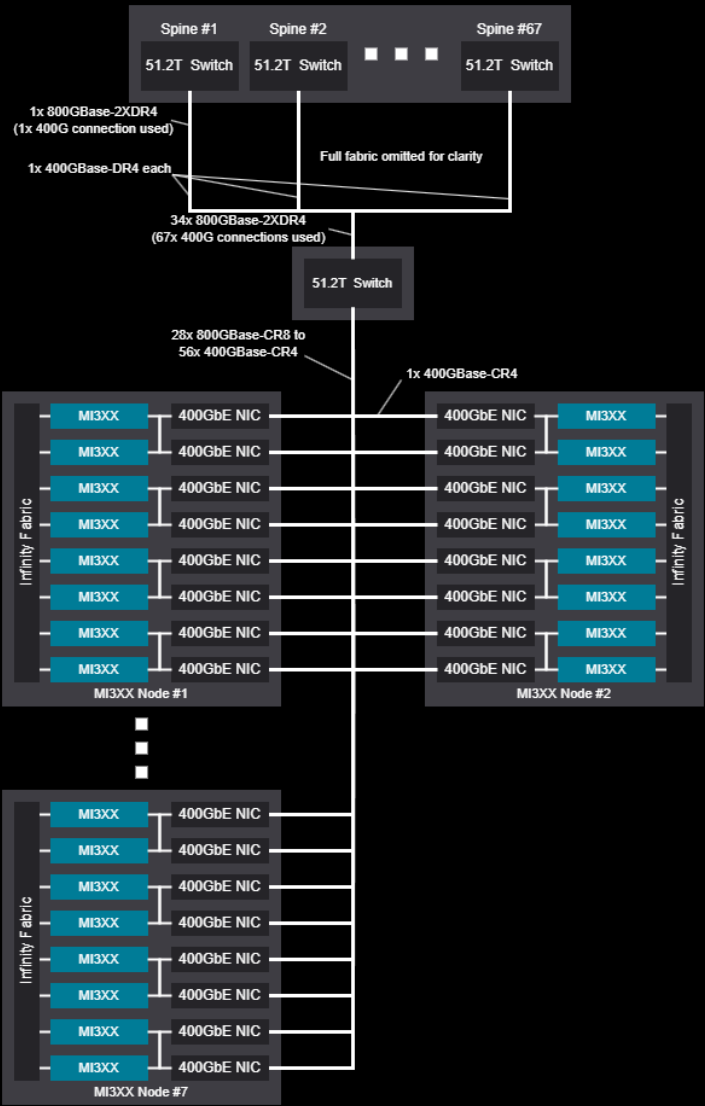
51.2T



Network Diagram - 4144 GPU (518 Nodes)

Tree Scalable Unit

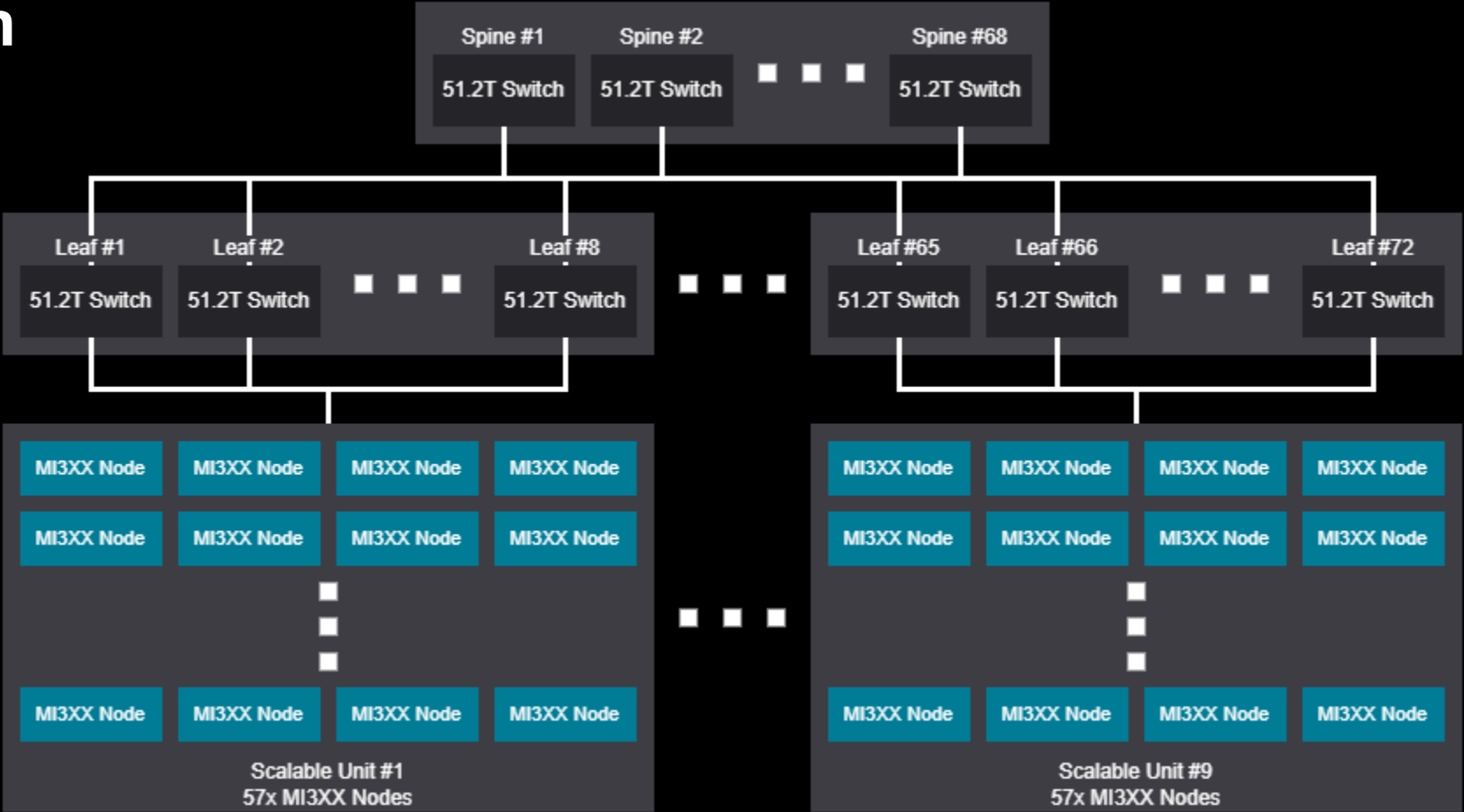
51.2T



Network Diagram - 4104 GPU (513 Nodes)

Rail Design

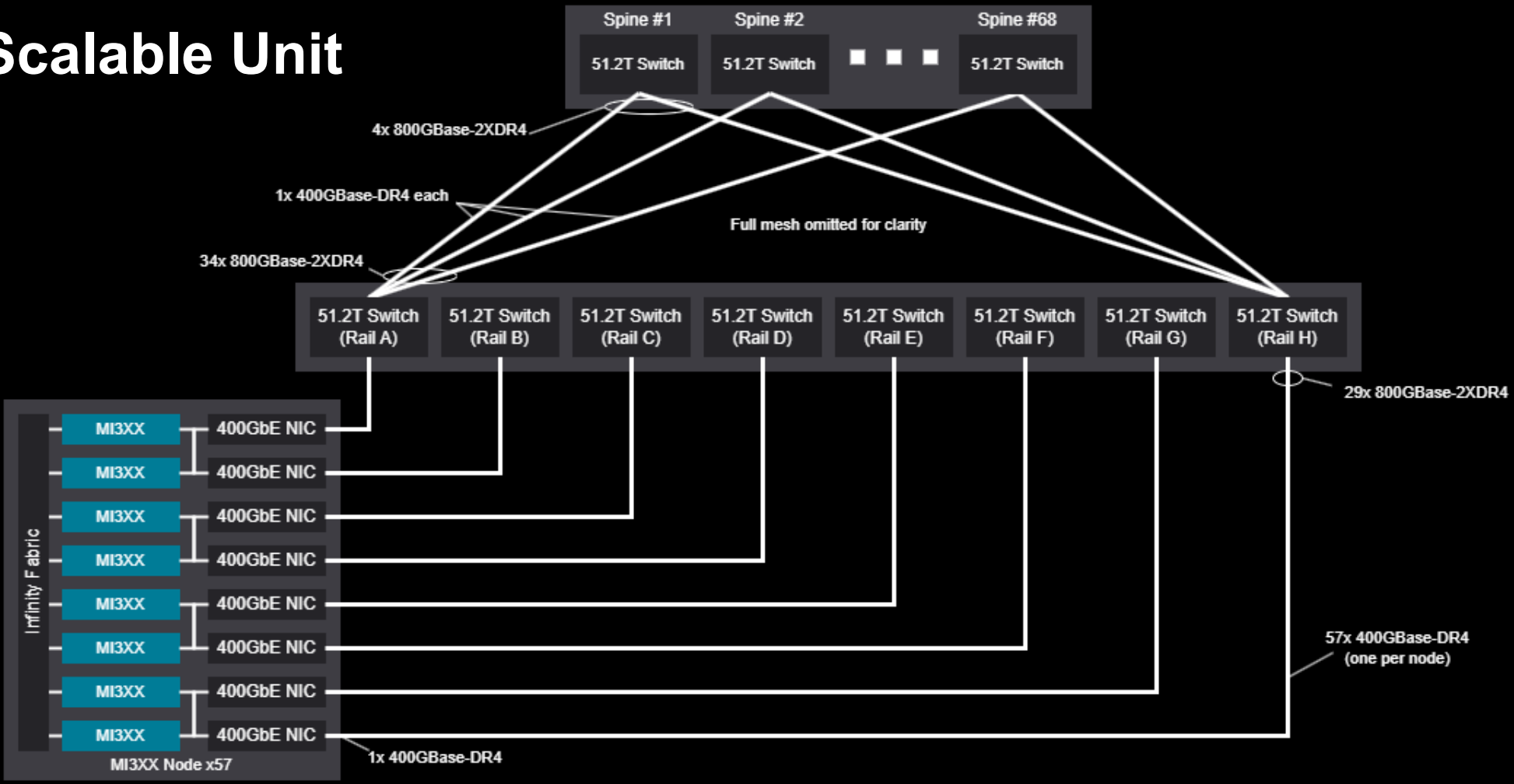
51.2T



Network Diagram - 4104 GPU (513 Nodes)

Rail Scalable Unit

51.2T

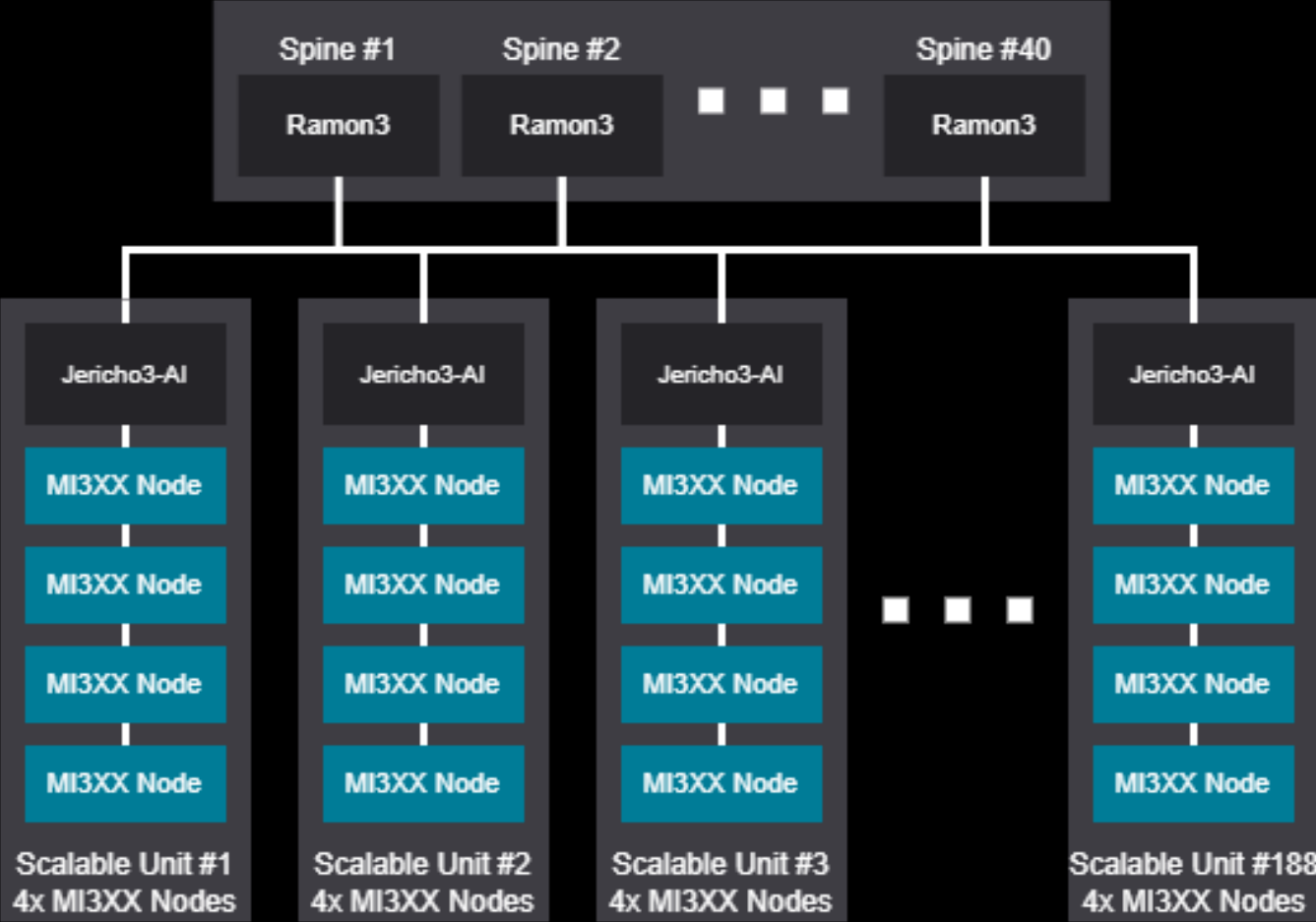


6K GPU Topology Design Examples – Scheduled Fabrics

Network Diagram - 6016 GPU (752 Nodes)

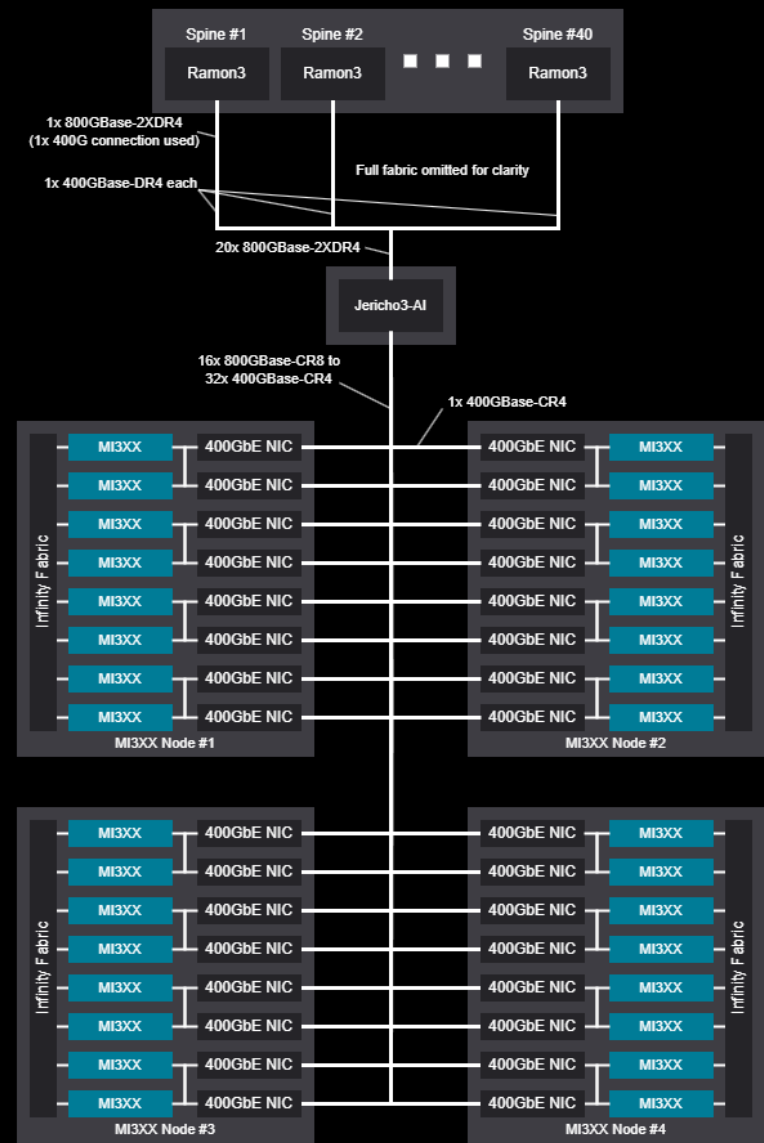
Tree Design

Jericho/Ramon



Network Diagram - 6016 GPU (752 Nodes)

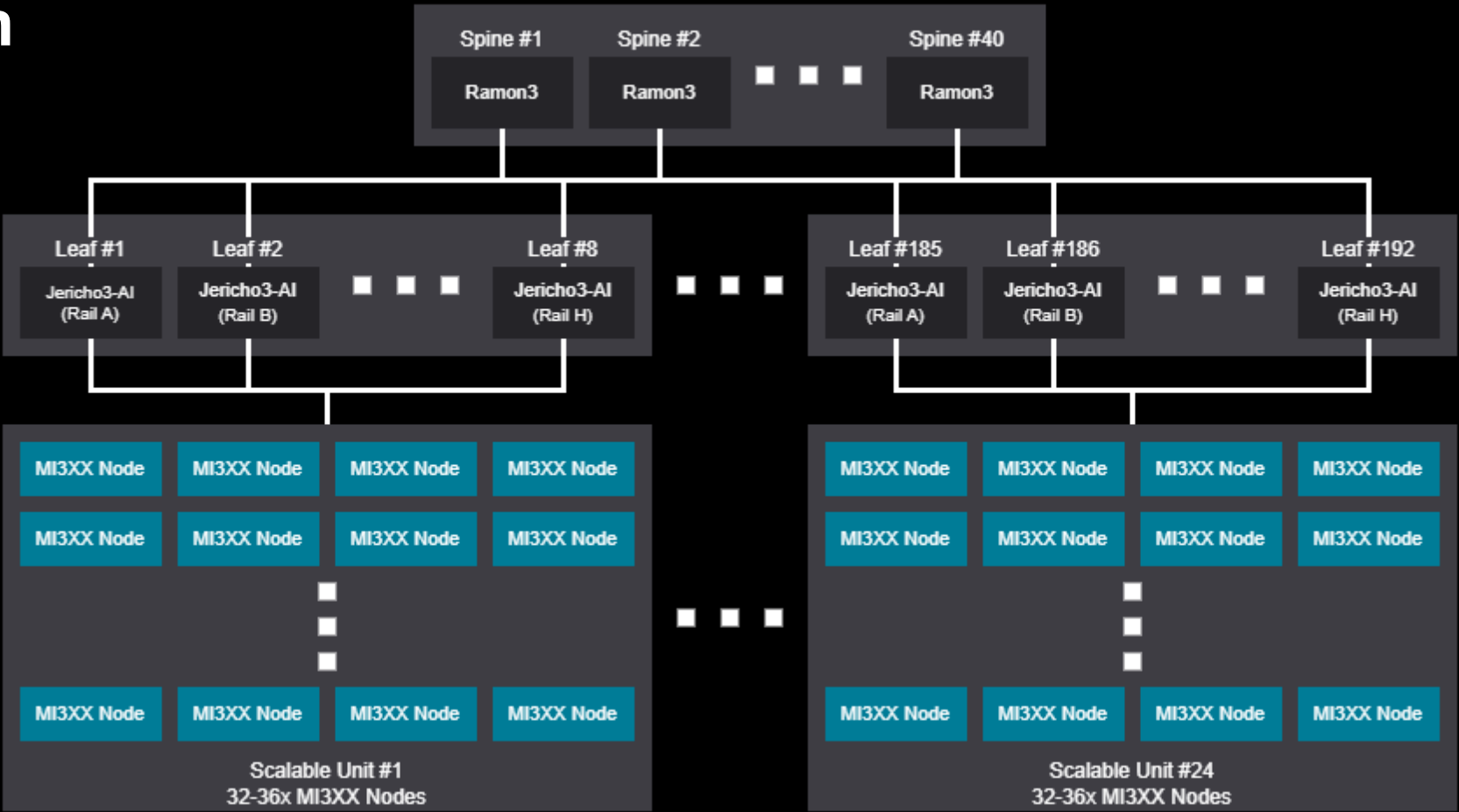
Tree Scalable Unit Jericho/Ramon



Network Diagram - 6144-6912 GPU (768-864 Nodes)

Rail Design

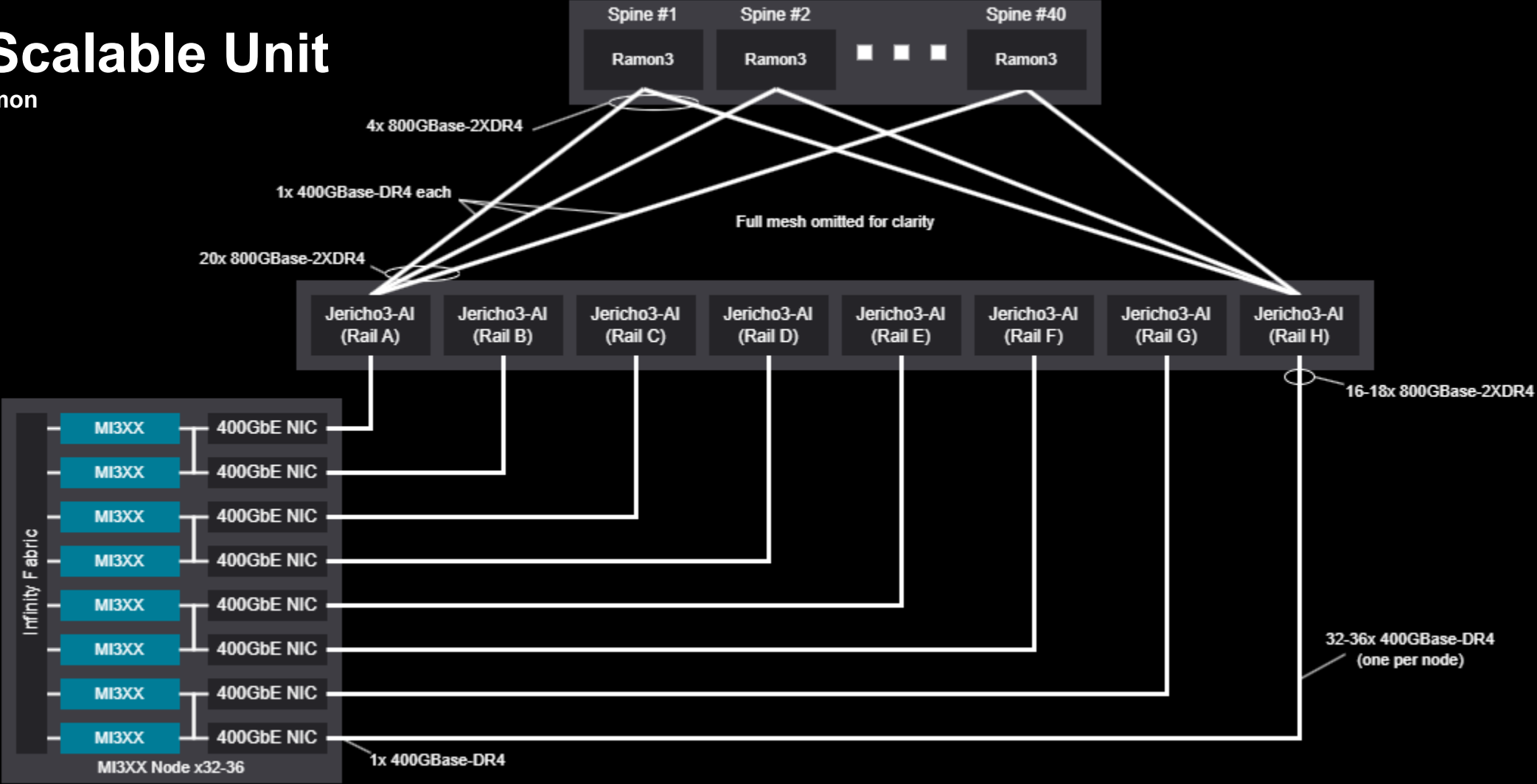
Jericho/Ramon



Network Diagram - 6144-6912 GPU (768-864 Nodes)

Rail Scalable Unit

Jericho/Ramon



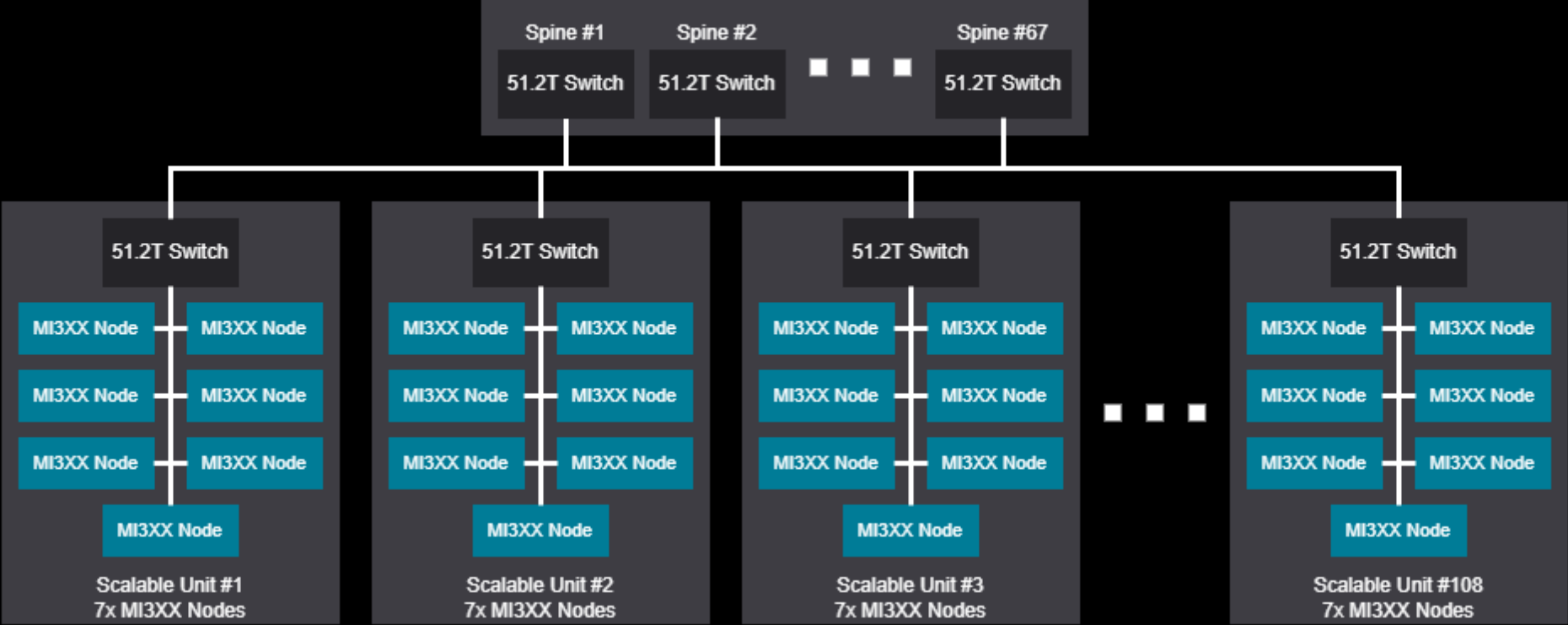
6K GPU Topology Design Examples

51.2T

Network Diagram - 6048 GPU (756 Nodes)

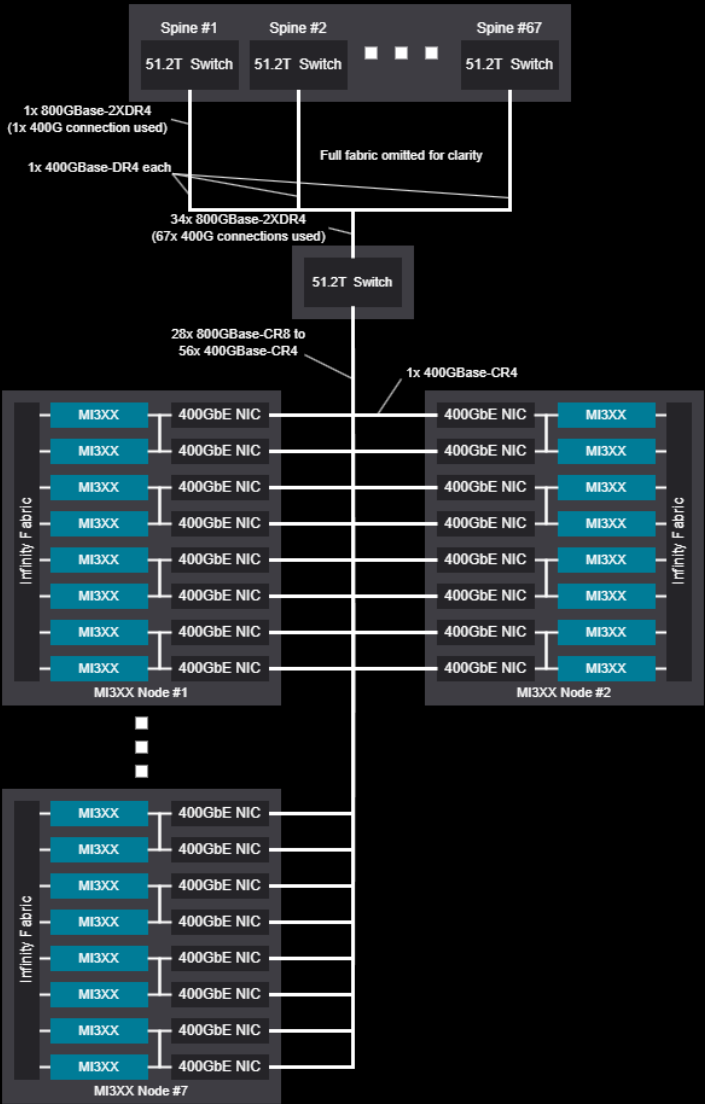
Tree Design

51.2T



Network Diagram - 6048 GPU (756 Nodes)

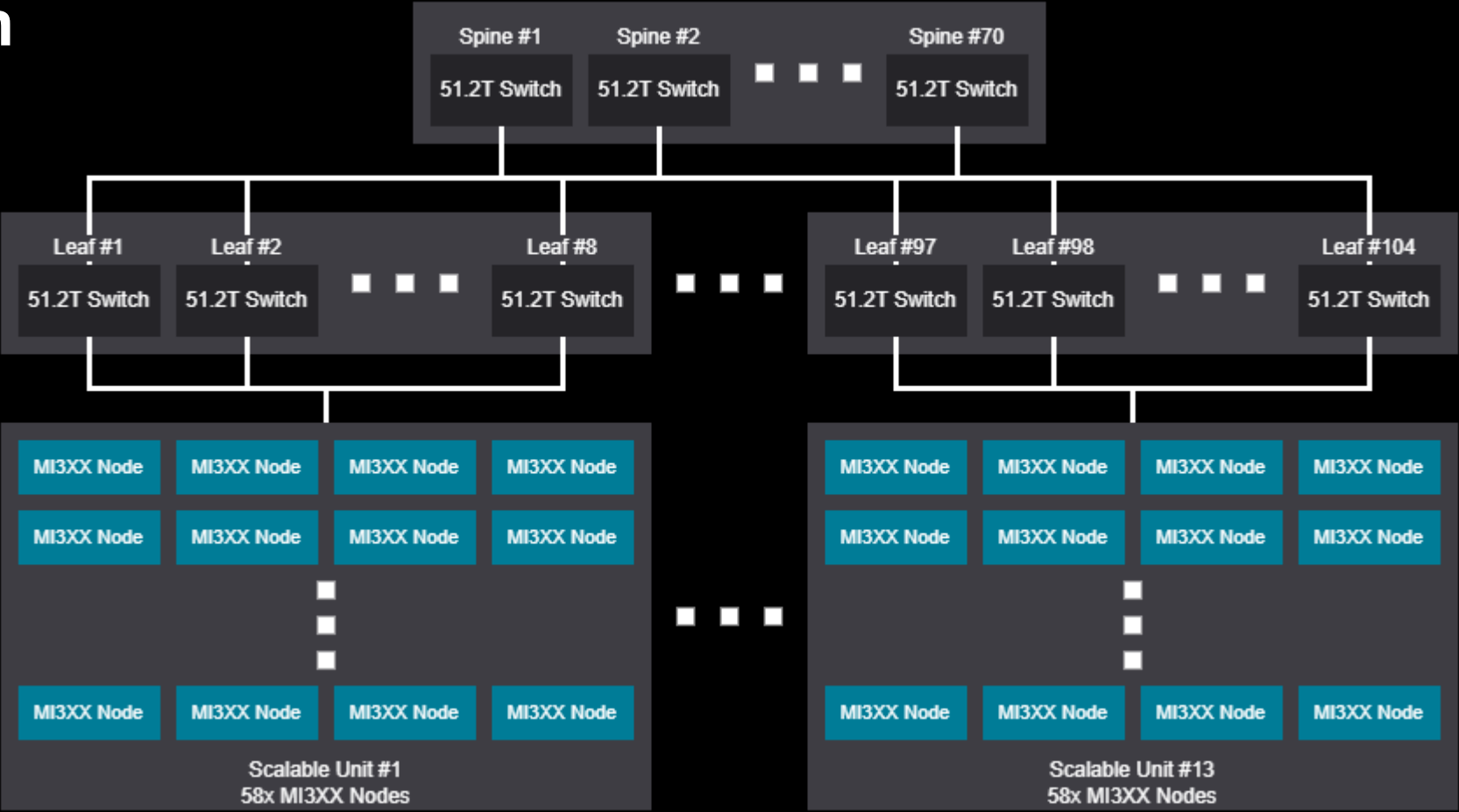
Tree Scalable Unit 51.2T



Network Diagram - 6032 GPU (754 Nodes)

Rail Design

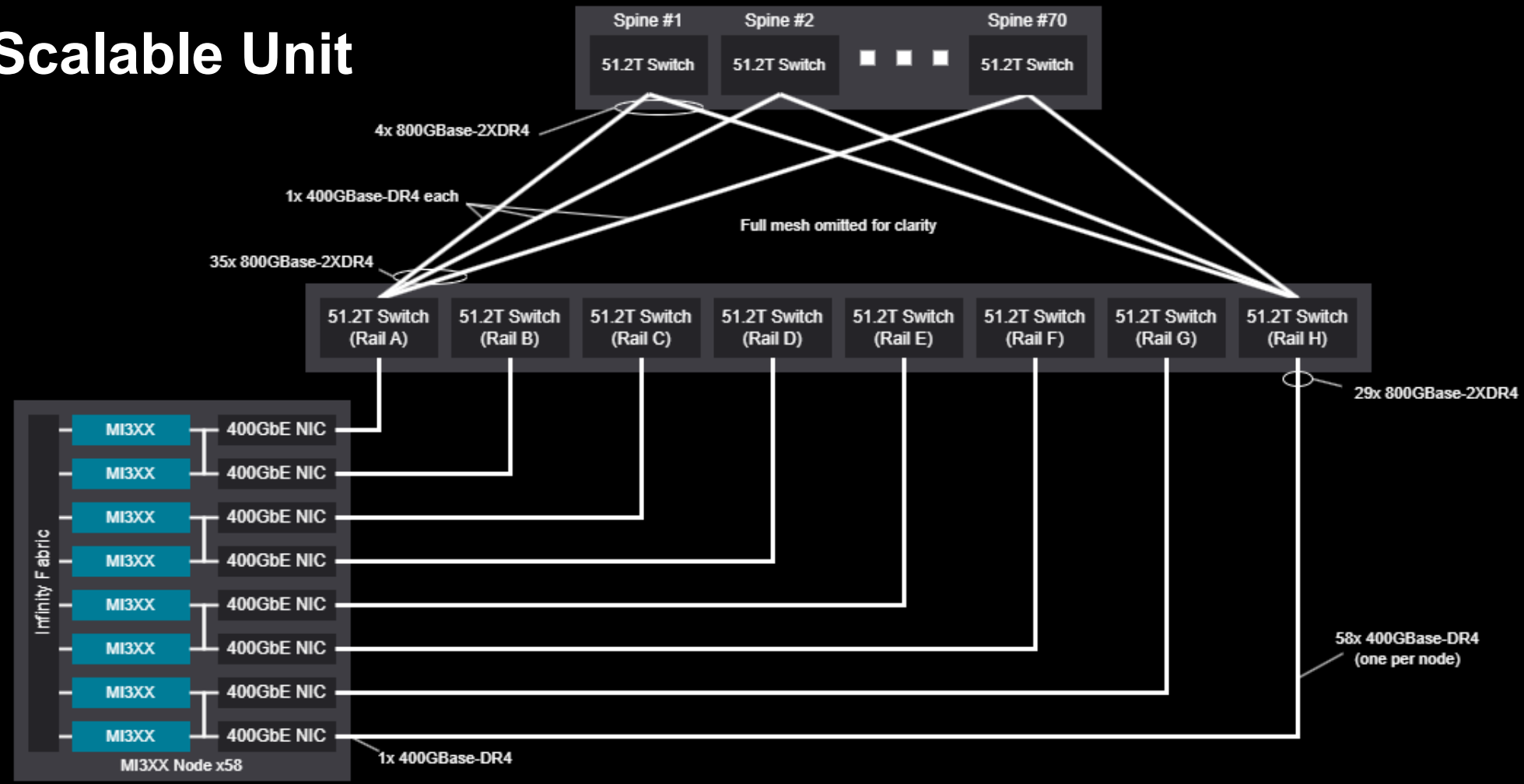
51.2T



Network Diagram - 6032 GPU (754 Nodes)

Rail Scalable Unit

51.2T

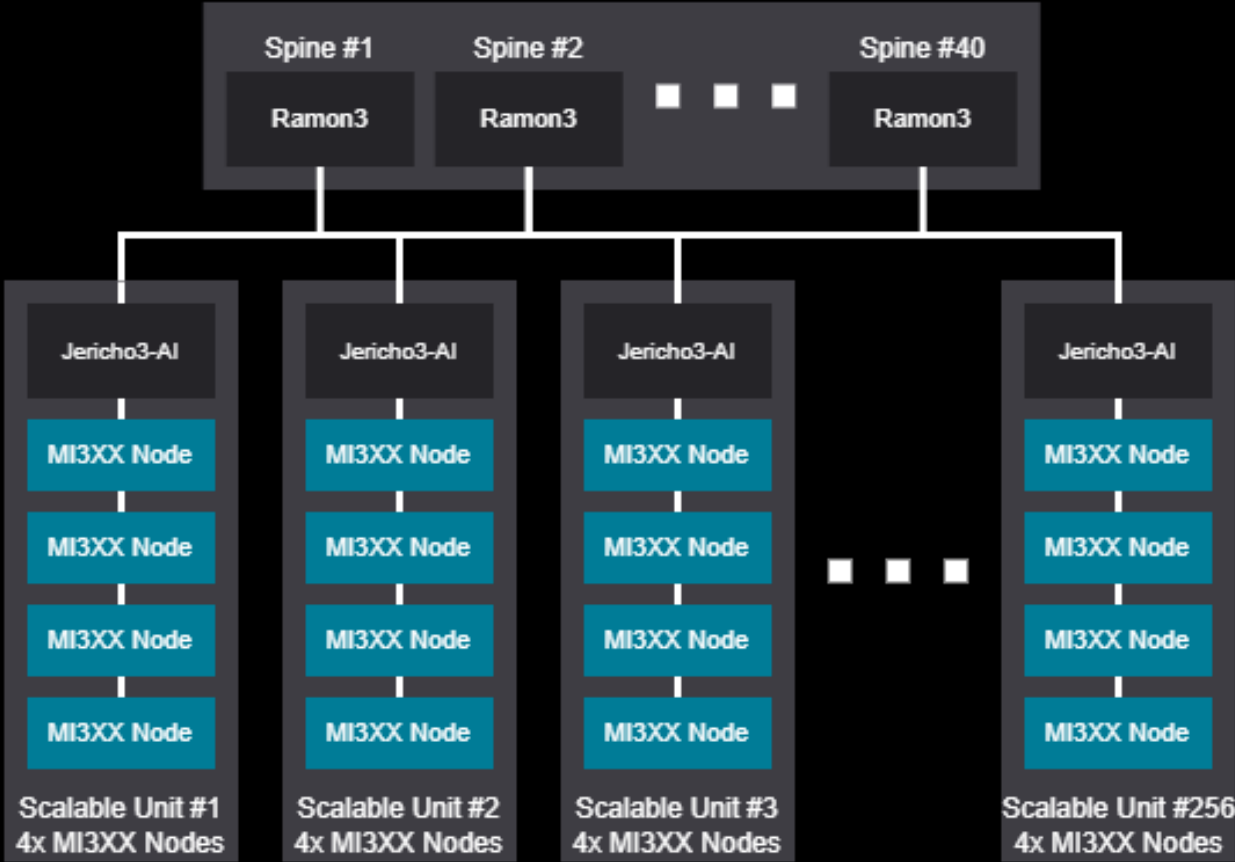


8K GPU Topology Design Examples – Scheduled Fabrics

Network Diagram - 8192 GPU (1024 Nodes)

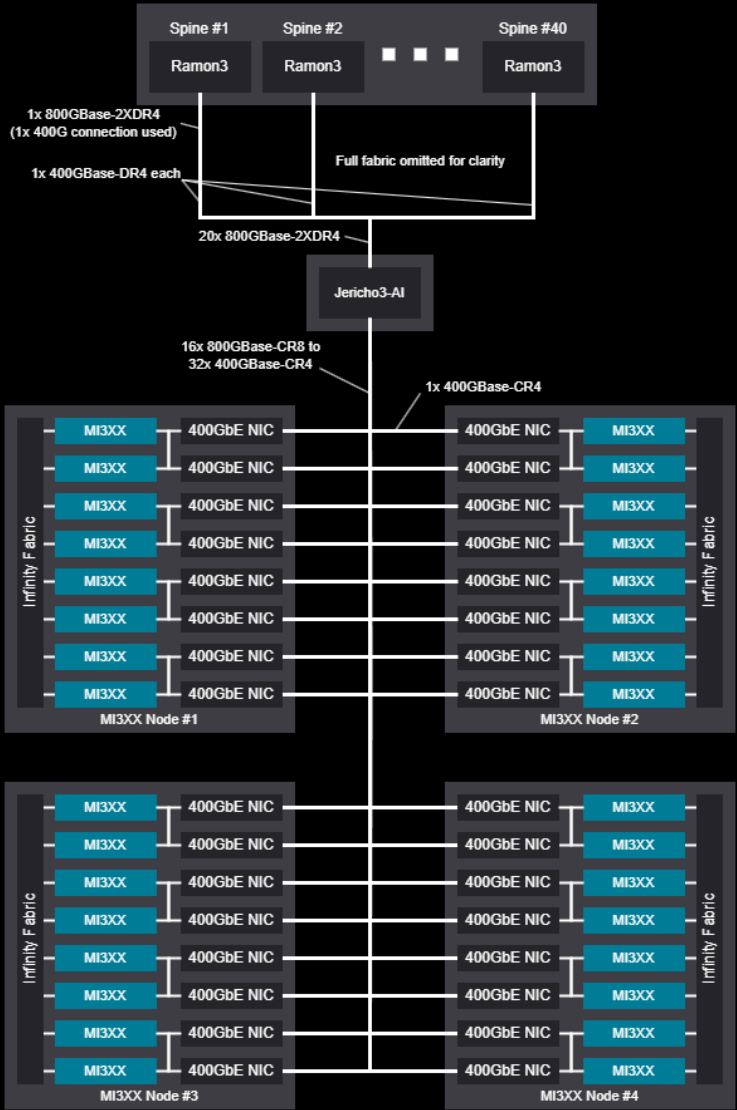
Tree Design

Jericho/Ramon



Network Diagram - 8192 GPU (1024 Nodes)

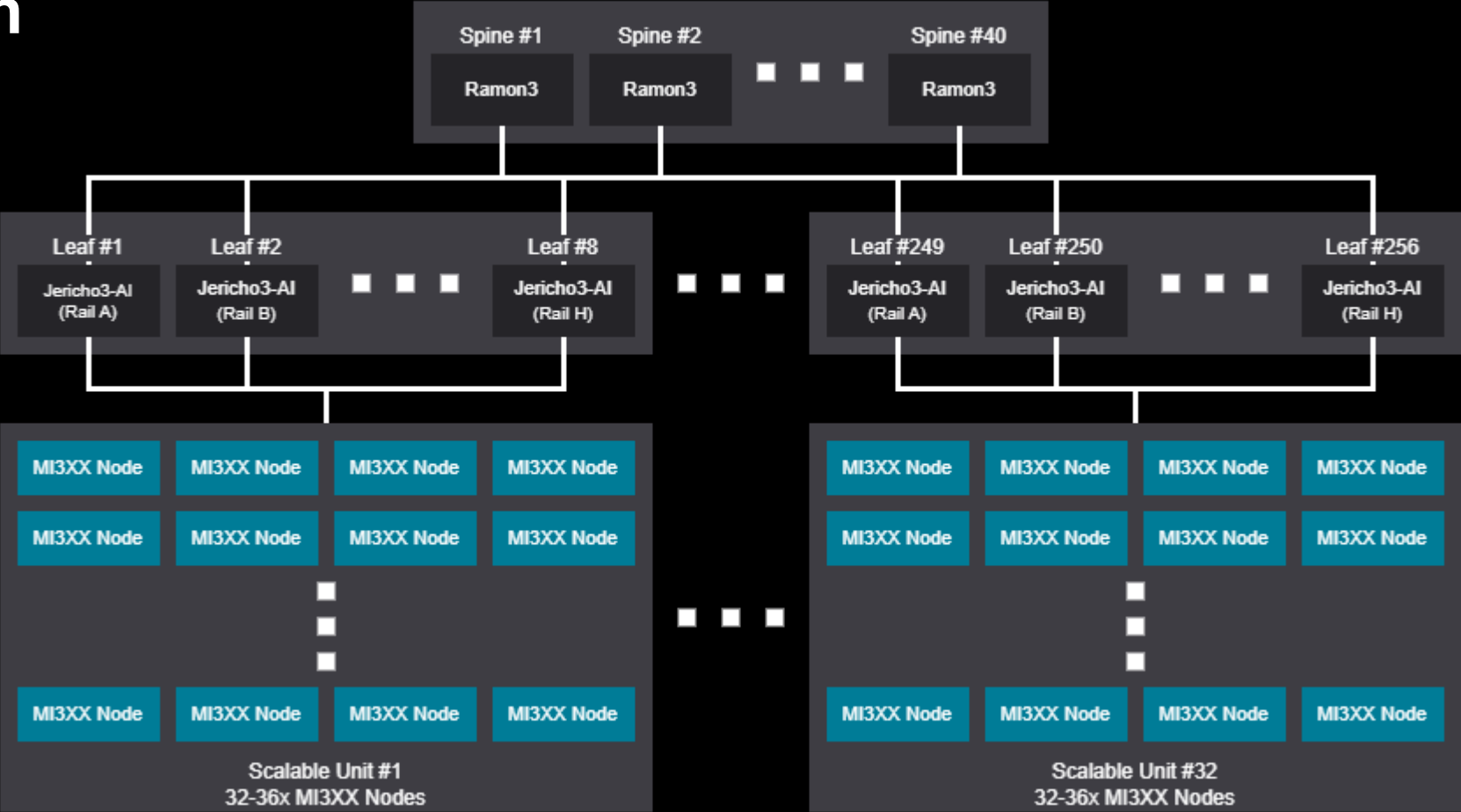
Tree Scalable Unit Jericho/Ramon



Network Diagram - 8192-9216 GPU (1024-1152 Nodes)

Rail Design

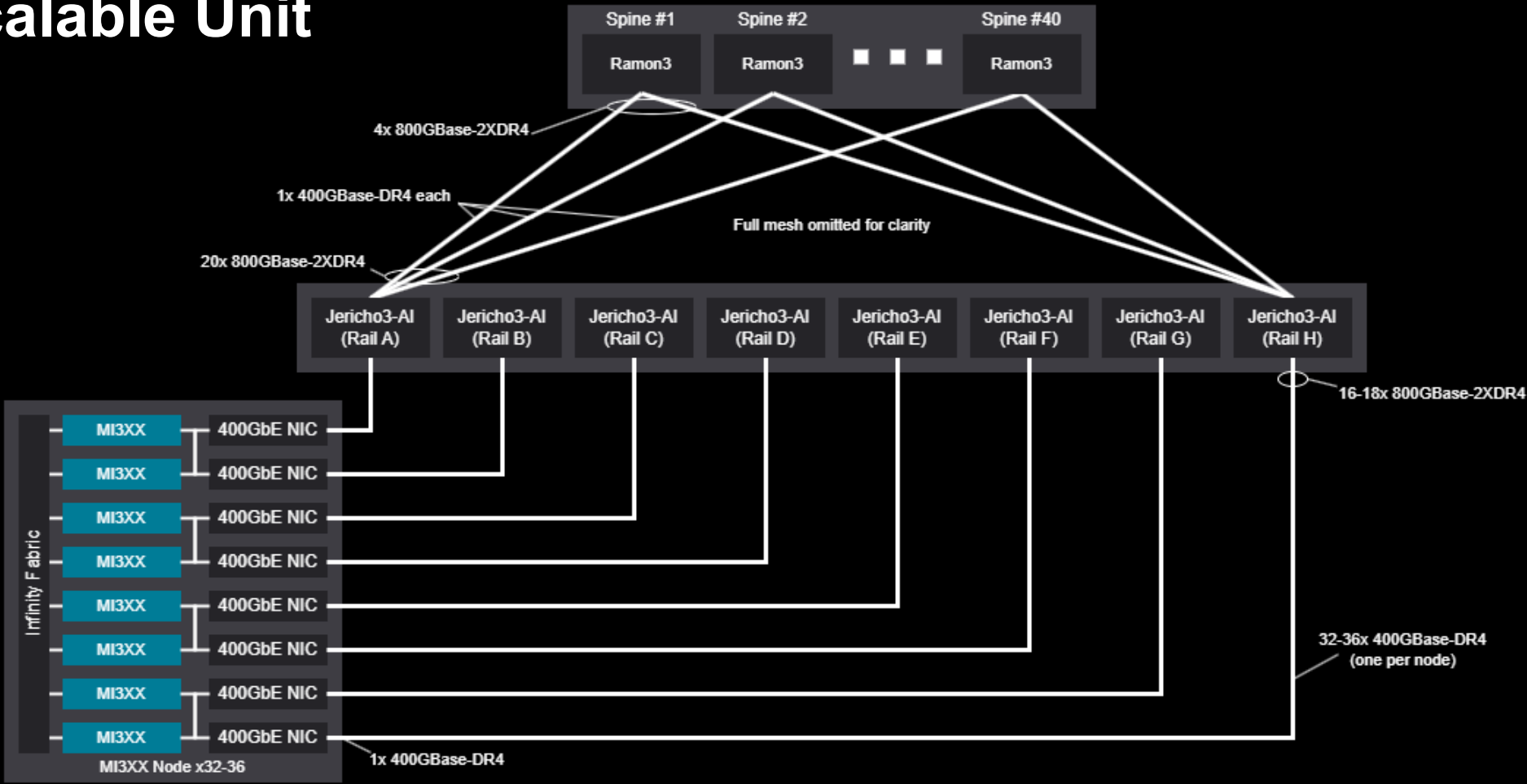
Jericho/Ramon



Network Diagram - 8192-9216 GPU (1024-1152 Nodes)

Rail Scalable Unit

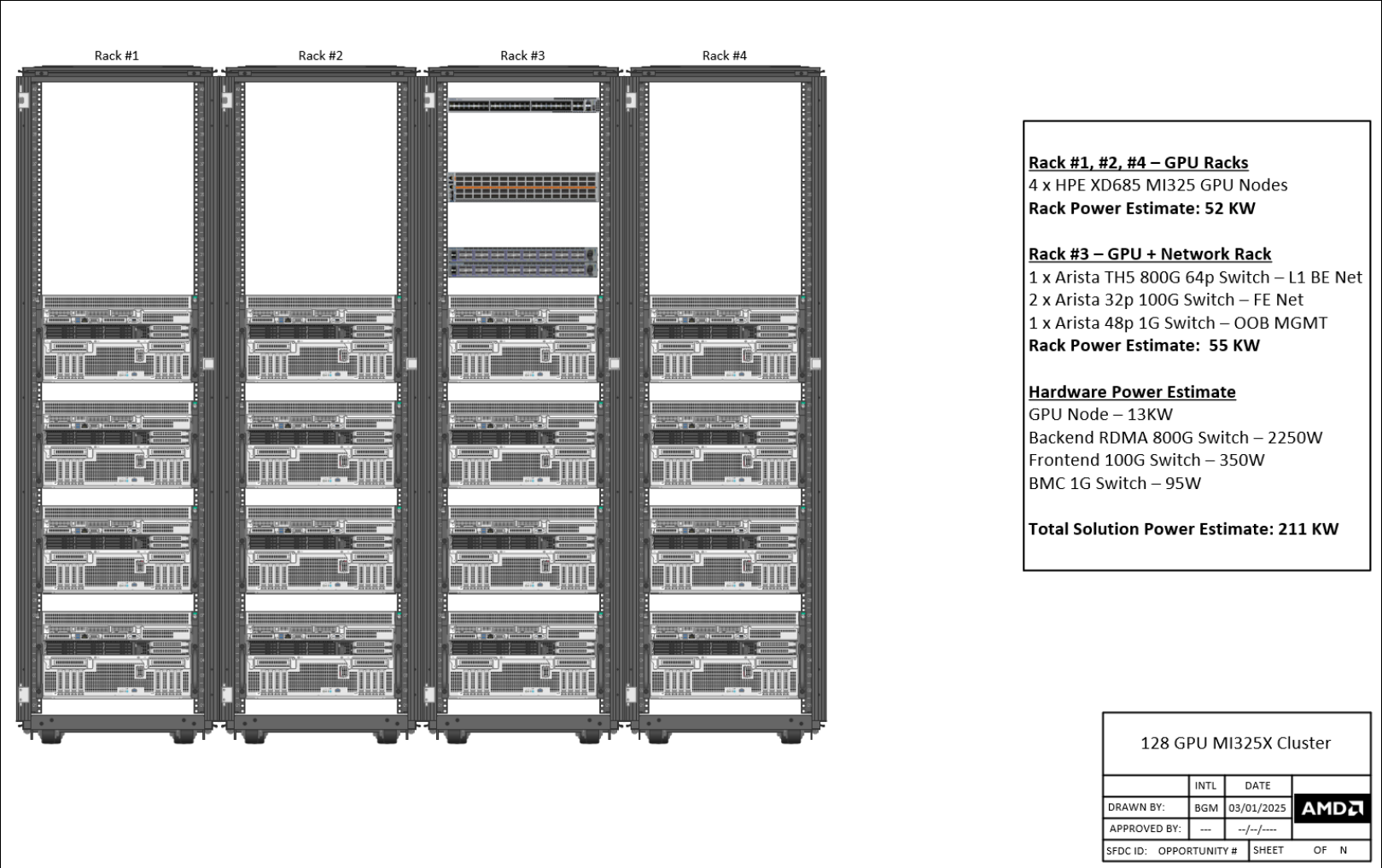
Jericho/Ramon





HPE – Arista Design – Sample Racks (POD)

Sample Rack Elevation - 128 GPU (16 Nodes) - Single Switch



Sample Rack Elevation - 512GPU (64 Node) POD- Rail-Optimized



Sample Rack Elevation - 1024GPU (128 Node) POD- Rail-Optimized

512 GPU – POD #1 – Rack #1-#18



Rack #1, #6, #11, #15-#22, #28-#31, #33-#36 – GPU Racks
4 x HPE X86 M325 GPU Nodes
Rack Power Estimate: 52 KW

Rack #5, #23 – L1 BE Network Rack
8 x Arista T15 800G 64p Switch – L1 BE Net
2 x Arista 64p 100G Switch – FE Net
1 x Arista 48p 1G Switch – OOB MGMT
Rack Power Estimate: 20 KW

Rack #14, #32 – L1 FE Network Rack
2 x Arista 64p 100G Switch – FE Net
1 x Arista 48p 1G Switch – OOB MGMT
Rack Power Estimate: 2 KW

Rack #XX – L2 Spine Network Rack
8 x Arista T15 800G 64p Switch – L2 BE Net
2 x Arista 64p 100G Switch – FE Net Spine
Rack Power Estimate: 20 KW

Hardware Power Estimate
M325 GPU Node – 13KW
Arista T15 800G 64p Switch – 2250W
Arista 100G 64p Switch – 927W
Arista 1G 48p Switch – 95W
Total Solution Power Estimate: 1,704 KW

512 GPU – POD #2 – Rack #19-#36



1024 GPU M325X Cluster POD	
DATE	2023
DESIGNED BY	AMDA
APPROVED BY	AMDA
DATE	2023

DISCLAIMER AND ATTRIBUTIONS

DISCLAIMER

The information contained herein is for informational purposes only, and is subject to change without notice. While every precaution has been taken in the preparation of this document, it may contain technical inaccuracies, omissions and typographical errors, and AMD is under no obligation to update or otherwise correct this information. Advanced Micro Devices, Inc. makes no representations or warranties with respect to the accuracy or completeness of the contents of this document, and assumes no liability of any kind, including the implied warranties of noninfringement, merchantability or fitness for particular purposes, with respect to the operation or use of AMD hardware, software or other products described herein. No license, including implied or arising by estoppel, to any intellectual property rights is granted by this document. Terms and limitations applicable to the purchase or use of AMD's products are as set forth in a signed agreement between the parties or in AMD's Standard Terms and Conditions of Sale. GD-18

COMPLIANCE WITH LAWS

Customer shall adhere to all applicable export laws and regulations including, without limitation, those administered by the U.S. Department of Commerce – Bureau of Industry and Security (U.S. Export Administration Regulations 15 CFR 730 et seq.) and those administered by the U.S. Department of State in accordance with the U.S. International Traffic in Arms Regulations (ITAR) set forth in Subchapter M, Title 22, Code of Federal Regulations, Parts 120 through 130 (22 CFR 120-130), as the same may be amended from time to time, and shall not export, re-export, resell, transfer, or disclose, directly or indirectly, any Products or technical data, or the direct product of any Products or technical data, to any proscribed person, entity, or country, or foreign national thereof, unless properly authorized by the U.S. government and/or any other applicable or relevant government or regulatory body, including the export authorities of all respective countries. For the avoidance of doubt, Customer shall not use Products in, or re-export Products to Belarus, Russia and the Donetsk (DNR) or Luhansk (LNR) regions of Ukraine, regardless of the applicable export laws and regulations. Customer shall impose upon its customers terms at least as restrictive as those contained in this Clause 14 with respect to any sale, distribution or export of Products.

© 2025 Advanced Micro Devices, Inc. All rights reserved. AMD, the AMD Arrow logo, AMD Instinct, AMD together we advance_, Infinity Fabric, ROCm, and combinations thereof are trademarks of Advanced Micro Devices, Inc. Amazon S3, Arista, Arista OSFP, APC, Broadcom, Ciena, Cisco, CloudVision, DataDirect Networks, Dell, DriveNEts, EOS, FS.com, Hammerspace, Hewlett-Packard Enterprise, IOS, Juniper, JUNOS, Lenovo, Linux, MTP, Netshelter, Nokia, Proliant, Pure Storage, Schneider Electric, SONiC, Super Micro Computer Inc, Tomahawk, Ubuntu, Vast Data, Weka, and other product names used in this publication are for identification purposes only and may be trademarks of their respective owners. Certain AMD technologies may require third-party enablement or activation. Supported features may vary by operating system. Please confirm with the system manufacturer for specific features. No technology or product can be completely secure.